Online Supplemental Appendix to: Nonparametric Spot Correlation Estimation and Fixed-k Inference Using High-Frequency Candlestick Data

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Abstract

This supplemental appendix is divided into three sections. Section SA contains all the remaining proofs of results in the main text. Section SB discusses a general minimaxity result in the quadratic consistent class of correlation estimators. Section SC presents additional simulation results.

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SA Proofs

Following Bollerslev et al. (2024) and Li et al. (2024), we shall assume that jumps are absent in intervals of order $O(k_n\Delta_n)$ near τ such that P is continuous on $\mathcal{I}_n \equiv \bigcup_{i=1}^{k_n} \mathcal{I}_{n,i}$, as the probability of observing a jump in this interval converges to zero as $k_n\Delta_n \to 0$. We also use the strengthened version of Assumption 1 by assuming that the conditions hold with $T_1 = \infty$ using a standard localization procedure (see the discussion in Li et al. (2024)). The positive constants K, K' are generic and may vary from line to line, independent of n. For a generic random variable X, we define $\|X\|_p = (\mathbb{E}[|X|^p])^{1/p}$ as its L_p-norm for $1 \leq p < \infty$.

Let $\mathcal{G} \subseteq \mathcal{F}$ denote a sub-σ-field of $\mathcal{F}$. As a preliminary, we shall give the definition of convergence in $\mathcal{G}$-conditional law in the vein of Li and Xiu (2016, Definition A1) and a useful lemma.

Definition 1. *Let (X_n) be a sequence of random variables defined on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, and let X denote a random variable defined on an extension of $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$. We say that X_n converges in $\mathcal{G}$-conditional law to X, written as $X_n \xrightarrow{\mathcal{L}|\mathcal{G}} X$, if for every bounded continuous function f, it holds that $\mathbb{E}[f(X_n) \,|\, \mathcal{G}] \xrightarrow{\mathbb{P}} \mathbb{E}[f(X) \,|\, \mathcal{G}]$, where the expectation is taken with respect to the extended probability space if necessary.*

The convergence in $\mathcal{G}$-conditional law clearly nests the mere convergence in law since $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ follows from $\mathbb{E}[f(X_n) \mid \mathcal{G}] \xrightarrow{\mathbb{P}} \mathbb{E}[f(X) \mid \mathcal{G}]$ using the boundedness of f. The following lemma is repeatedly used in the proof, with $\mathcal{G}$ taken to be $\mathcal{F}_\tau$:

Lemma 1. *For two sequences of random variables (X_n) and (Y_n) and a random variable X with law $\mathbb{P}_X$, if $X_n \xrightarrow{\mathcal{L}|\mathcal{G}} X$ and $|Y_n - X_n| \xrightarrow{\mathbb{P}} 0$, then*

(i) (Continuous mapping.) For any Borel measurable $\mathbb{P}_X$-almost everywhere continuous function f, it holds that $|f(Y_n) - f(X_n)| \xrightarrow{\mathbb{P}} 0$.

(ii) (Asymptotic equivalence.) $Y_n \xrightarrow{\mathcal{L}|\mathcal{G}} X$.

Proof. For (i), by $X_n \xrightarrow{\mathcal{L}} X$ and $|X_n - Y_n| \xrightarrow{\mathbb{P}} 0$, we have the joint convergence $(X_n, Y_n) \Rightarrow (X, X)$ by Slutsky's Theorem. Then $(f(X_n), f(Y_n)) \Rightarrow (f(X), f(X))$ and hence $|f(X_n) - f(Y_n)| \xrightarrow{\mathcal{L}} 0$ follows by the continuous mapping theorem. But since the limit is degenerate, we conclude that $|f(X_n) - f(Y_n)| \xrightarrow{\mathbb{P}} 0$.

For (ii), take any bounded continuous function φ, it holds from (i) that $|\varphi(Y_n) - \varphi(X_n)| \xrightarrow{\mathbb{P}} 0$. With the boundedness of φ, we further deduce $\|\varphi(Y_n) - \varphi(X_n)\|_1 \to 0$. The required claim follows from

$$\begin{aligned}\|\mathbb{E}[\varphi(Y_n) \mid \mathcal{G}] - \mathbb{E}[\varphi(X) \mid \mathcal{G}]\|_1 &\leq \|\mathbb{E}[\varphi(Y_n) - \varphi(X_n) \mid \mathcal{G}]\|_1 + \|\mathbb{E}[\varphi(X_n) \mid \mathcal{G}] - \mathbb{E}[\varphi(X) \mid \mathcal{G}]\|_1 \\ &\leq \|\varphi(Y_n) - \varphi(X_n)\|_1 + \|\mathbb{E}[\varphi(X_n) \mid \mathcal{G}] - \mathbb{E}[\varphi(X) \mid \mathcal{G}]\|_1 \to 0,\end{aligned}$$

where the last estimate is by the assumption that $X_n \xrightarrow{\mathcal{L}|\mathcal{G}} X$. The above L_1 convergence implies that $\mathbb{E}[\varphi(Y_n) \mid \mathcal{G}] \xrightarrow{\mathbb{P}} \mathbb{E}[\varphi(X) \mid \mathcal{G}]$, which completes the proof. □

We shall prove some analytical properties for the function $g_\lambda(\cdot)$ defined in (8).

Lemma 2. *For every $\lambda \in [0,1]$, the function $g_\lambda : [-1,1] \to [-1,1]$ is an odd, continuous, and strictly increasing bijection. Moreover, $g_\lambda(\cdot)$ is differentiable on $(-1,1)$ with $g'_\lambda(\cdot)$ continuous on $(-1,1)$ and satisfies*

$$0 < \frac{1 - 8/\pi^2}{3 - 4\log 2} \leq g'_\lambda(\rho) \leq \frac{1}{3 - 4\log 2}, \qquad \rho \in (-1,1).$$

Consequently, $g_\lambda^{-1}(\cdot)$ exists, is Lipschitz and continuously differentiable on $(-1,1)$, with

$$0 < 3 - 4\log 2 \leq (g_\lambda^{-1})'(u) = \frac{1}{g'_\lambda(g_\lambda^{-1}(u))} \leq \frac{3 - 4\log 2}{1 - 8/\pi^2}, \qquad u \in (-1,1).$$

Proof of Lemma 2. Let $B = (B_t)_{0 \leq t \leq 1}$ be a standard Brownian motion and define the functionals $C(B) \equiv B_1$, $H(B) \equiv \sup_{0 \leq t \leq 1} B_t$, $L(B) \equiv \inf_{0 \leq t \leq 1} B_t$, and $A(B) \equiv H(B) + L(B) - C(B)$. One can verify that:

$$C(-B) = -C(B), \quad H(-B) = -L(B), \quad L(-B) = -H(B),$$

so we have $A(-B) = -A(B)$, hence $A(B)$ is odd in B with $\mathbb{E}[A(B)] = 0$.

For $\rho \in [-1,1]$, let $B^\perp$ be an independent copy of B, and define

$$B^\rho \equiv \rho B + \sqrt{1-\rho^2} B^\perp, \tag{SA.1}$$

so that (B, B^ρ) is a bivariate Brownian motion with correlation ρ. Define $g_A(\rho) \equiv \mathbb{E}[A(B)A(B^\rho)]$, and note that $g_A(\rho) = 2(c(\rho) - c(-\rho)) - \rho$. With this notation, we can rewrite $g_\lambda(\rho)$ as

$$g_\lambda(\rho) = \frac{(1-\lambda)\rho + \lambda g_A(\rho)}{(1-\lambda) + \lambda(3 - 4\log 2)}. \tag{SA.2}$$

We shall first prove oddness and continuity of $g_\lambda(\rho)$ on $[-1,1]$, which immediately follows from the oddness and continuity of $g_A(\rho)$. First, notice that $(B, B^{-\rho}) \stackrel{d}{=} (-B, B^\rho)$, and therefore, for all $\rho \in [-1,1]$:

$$-g_A(\rho) = \mathbb{E}[-A(B)A(B^\rho)] = \mathbb{E}[A(-B)A(B^\rho)] = \mathbb{E}[A(B)A(B^{-\rho})] = g_A(-\rho),$$

so g_A is odd. To prove the continuity of g_A, take any sequence $\rho_n \to \rho$ on $[-1,1]$. The construction of B^ρ in SA.1 implies that $\sup_{0 \leq t \leq 1} |B_t^{\rho_n} - B_t^\rho| \to 0$ almost surely, so B^ρ is continuous in ρ under the sup-norm. The candlestick maps $B \mapsto C(B)$, $B \mapsto H(B)$, $B \mapsto L(B)$, and hence $B \mapsto A(B)$ are continuous under the sup-norm, thus $A(B)A(B^{\rho_n}) \to A(B)A(B^\rho)$, almost surely. Now, write $M \equiv \sup_{0 \leq t \leq 1} |B_t| + \sup_{0 \leq t \leq 1} |B_t^\perp|$, and it is well-known that Brownian suprema have finite moments of all orders, i.e., $\mathbb{E}[M^k] < \infty, \forall k \in \mathbb{N}$. Uniformly over $\rho \in [-1,1]$, it holds that

$$|C(B^\rho)| \leq M, \quad |H(B^\rho)| \leq M, \quad |L(B^\rho)| \leq M,$$

and hence $|A(B^\rho)| \le 3M$. This further implies that $|A(B)A(B^\rho)| \le 9M^2$. Since $\mathbb{E}[M^2] < \infty$, by the dominated convergence theorem, we conclude that

$$g_A(\rho_n) = \mathbb{E}[A(B)A(B^{\rho_n})] \to \mathbb{E}[A(B)A(B^\rho)] = g_A(\rho),$$

proving the continuity of g_A on $[-1, 1]$.

To provide estimates for the derivative of g_A, we use Malliavin calculus techniques. On the Wiener space over $[0, 1]$, let $\mathcal{H} = L^2([0, 1])$. To highlight the dimension, denote $\|f\|_{\mathcal{H}(q)} \equiv (\int_{[0,1]^q} f^2(x)dx)^{1/2}$ for $f : [0, 1]^q \to \mathbb{R}$ defined on q-dimensional cube. The Malliavin derivative D is first defined on smooth cylindrical random variables and then closed as an operator from $L^2(\Omega)$ into $L^2(\Omega; \mathcal{H})$. The first-order Malliavin–Sobolev space $\mathbb{D}^{1,2}$ is the domain of this closed operator under the norm

$$\|F\|_{1,2}^2 = \mathbb{E}[F^2] + \mathbb{E}[\|DF\|_{\mathcal{H}(1)}^2],$$

where D denotes the Malliavin derivative as an $\mathcal{H}$-valued operator, while D_sF denotes its coordinate at time s, so that $DF = (D_sF)_{0\le s\le 1}$, see Nualart (2006, Section 1.2).

Let τ_H and τ_L denote the almost surely unique times at which B attains its maximum and minimum on $[0, 1]$. The Malliavin derivative identity for the Brownian maximum gives $H \in \mathbb{D}^{1,2}$ and $D_sH = \mathbb{1}_{\{s\le\tau_H\}}$, see Nualart (2006, Exercise 1.2.11 and Proposition 2.1.10). Applying the same result to $-B$ yields $L \in \mathbb{D}^{1,2}$ and $D_sL = \mathbb{1}_{\{s\le\tau_L\}}$. Also, $C(B) = B_1 \in \mathbb{D}^{1,2}$ and $D_sC \equiv 1$. Therefore,

$$A = H + L - C \in \mathbb{D}^{1,2}, \quad D_sA = \mathbb{1}_{\{s\le\tau_H\}} + \mathbb{1}_{\{s\le\tau_L\}} - 1.$$

In particular, $|D_sA| \le 1$ for all $s \in [0, 1]$, so $\mathbb{E}[\|DA\|_{\mathcal{H}(1)}^2] \le 1$.

Because $A \in \mathbb{D}^{1,2} \subset L^2(\Omega)$ and A is measurable with respect to the Brownian path, the Wiener–Itô chaos decomposition gives a unique expansion $A = \mathbb{E}[A] + \sum_{q=1}^\infty I_q(f_q)$, where $f_q \in L^2([0, 1]^q)$ and $I_q(f_q)$ denotes the q-th multiple Wiener integral, see Nualart (2006, Section 1.1). Recall that A is odd with $\mathbb{E}[A] = 0$, and the reflection $B \mapsto -B$ changes the sign of $I_q(f_q)$ by the factor $(-1)^q$. By uniqueness of the Wiener–Itô expansion, all even-order chaos kernels must vanish. Hence

$$A = \sum_{q\in\mathbb{N}_{\text{odd}}} I_q(f_q).$$

By the standard derivative formula for Wiener chaos expansions, we further have $\mathbb{E}[\|DA\|_{\mathcal{H}(1)}^2] = \sum_{q\in\mathbb{N}_{\text{odd}}} qq!\|f_q\|_{\mathcal{H}(q)}^2 \le 1$, see Nualart (2006, Proposition 1.2.2).

We compute the first chaos coefficient explicitly. Since $D_sI_q(f_q) = qI_{q-1}(f_q(\cdot, s))$, taking expectations in the derivative expansion gives $f_1(s) = \mathbb{E}[D_sA]$. By the arcsine law for the Brownian maximizer, $\mathbb{P}(\tau_H \le s) = \frac{2}{\pi}\arcsin\sqrt{s}$, for $s \in [0, 1]$, see Mörters and Peres (2010, Theorem 5.26). The same formula holds for τ_L, since τ_L is the maximizer of $-B$. Hence $f_1(s) = \mathbb{P}(\tau_H \ge s) + \mathbb{P}(\tau_L \ge s) - 1 = 1 - \frac{4}{\pi}\arcsin\sqrt{s}$. Consequently,

$$\|f_1\|_{\mathcal{H}(1)}^2 = \int_0^1 \Big(1 - \frac{4}{\pi}\arcsin\sqrt{s}\Big)^2 ds = 1 - \frac{8}{\pi^2}.$$

We now use the chaos expansion to differentiate g_A. By Mehler's formula (Nualart, 2006, Section 1.4, Equation 1.67) for the Ornstein–Uhlenbeck semigroup $\{T_t, t \geq 0\}$, for $\rho = e^{-t} \in (0, 1]$ and a square-integrable Brownian functional F,

$$T_t F(B) = \mathbb{E}[F(\rho B + \sqrt{1-\rho^2} B^{\perp}) \mid B] = \mathbb{E}[F(B^{\rho}) \mid B].$$

Taking $F(B) = I_q^B(f_q)$, and using the fact that the q-th Wiener chaos is an eigenspace of T_t with eigenvalue e^{-qt}, we obtain for $\rho \in (0, 1]$,

$$\mathbb{E}[I_q^{B^{\rho}}(f_q) \mid B] = T_t I_q^B(f_q) = e^{-qt} I_q^B(f_q) = \rho^q I_q^B(f_q). \tag{SA.3}$$

For $\rho \in [-1, 0)$, we may take $r = -\rho = e^{-t}$ and apply the previous argument with $-B$ in place of B, which gives $I_q^{-B}(f_q) = (-1)^q I_q^B(f_q)$ and hence $\mathbb{E}[I_q^{B^{\rho}}(f_q) \mid B] = r^q (-1)^q I_q^B(f_q) = \rho^q I_q^B(f_q)$. For $\rho = 0$, the identity in (SA.3) follows from independence, since $\mathbb{E}[I_q^{B^{\perp}}(f_q) \mid B] = 0$, for $q \geq 1$. So (SA.3) holds for all $\rho \in [-1, 1]$. Therefore, by (SA.3), the law of iterated expectation, and the orthogonality of Wiener chaoses, we have

$$\mathbb{E}[I_p^B(f_p) I_q^{B^{\rho}}(f_q)] = \rho^q \mathbb{E}[I_p^B(f_p) I_q^B(f_q)] = \mathbb{1}_{\{p=q\}} \rho^q q! \|f_q\|^2_{\mathcal{H}(q)},$$

for all $\rho \in [-1, 1]$. Plugging the chaos expansion of A into the expression of g_A and simplifying using the above relation yields

$$g_A(\rho) = \mathbb{E}[A(B) A(B^{\rho})] = \sum_{q \in \mathbb{N}_{\text{odd}}} \rho^q q! \|f_q\|^2_{\mathcal{H}(q)}.$$

The series is absolutely and uniformly convergent on $[-1, 1]$, since

$$\sum_{q \in \mathbb{N}_{\text{odd}}} |\rho^q| q! \|f_q\|^2_{\mathcal{H}(q)} \leq \sum_{q \in \mathbb{N}_{\text{odd}}} q! \|f_q\|^2_{\mathcal{H}(q)} = \mathbb{E}[A(B)^2] = 3 - 4 \log 2,$$

where the last estimate follows from Popov (2016). Moreover, since

$$\sum_{q \in \mathbb{N}_{\text{odd}}} q |\rho^{q-1}| q! \|f_q\|^2_{\mathcal{H}(q)} \leq \sum_{q \in \mathbb{N}_{\text{odd}}} q q! \|f_q\|^2_{\mathcal{H}(q)} = \mathbb{E}[\|DA\|^2_{\mathcal{H}(1)}] \leq 1,$$

the ρ-derivative series is also absolutely and uniformly convergent on $[-1, 1]$. Hence g_A is differentiable on $(-1, 1)$, with

$$g_A'(\rho) = \sum_{q \in \mathbb{N}_{\text{odd}}} q \rho^{q-1} q! \|f_q\|^2_{\mathcal{H}(q)} \leq 1. \tag{SA.4}$$

Since its partial sums are continuous functions of ρ, the uniform limit theorem implies that g_A' is continuous. Consequently, g_A' is continuous on $(-1, 1)$. For odd q, $q-1$ is even, and therefore $\rho^{q-1} \geq 0$ for every $\rho \in [-1, 1]$. Thus we also have

$$g_A'(\rho) \geq g_A'(0) = 1! \|f_1\|^2_{\mathcal{H}(1)} = 1 - \frac{8}{\pi^2}, \tag{SA.5}$$

and it is worth noting that $c'(0) = \big(g_A'(0)+1\big)/4 = 1/2 - 2/\pi^2$. By (SA.2) we have

$$g_\lambda'(\rho) = \frac{(1-\lambda)+\lambda g_A'(\rho)}{(1-\lambda)+\lambda(3-4\log 2)}.$$

The continuity of g_A' on $(-1,1)$ thus implies the continuity of g_λ' on $(-1,1)$. Moreover, we conclude from (SA.4) that

$$g_\lambda'(\rho) \le \frac{(1-\lambda)+\lambda}{(1-\lambda)+\lambda(3-4\log 2)} \le \frac{1}{3-4\log 2}.$$

Also, it follows from (SA.5) that

$$g_\lambda'(\rho) \ge \frac{(1-\lambda)+\lambda(1-8/\pi^2)}{(1-\lambda)+\lambda(3-4\log 2)}.$$

Since $1-8/\pi^2 < 3-4\log 2$, the right-hand side is decreasing in $\lambda \in [0,1]$, and hence is bounded below by its value at $\lambda = 1$. Thus

$$g_\lambda'(\rho) \ge \frac{1-8/\pi^2}{3-4\log 2} \equiv K_g > 0.$$

The above result shows that g_λ' is continuous on $(-1,1)$, and hence $g_\lambda \in C^1([-1,1])$. Moreover, since $g_\lambda'(\rho) \ge K_g > 0$, g_λ is strictly increasing. At $\rho = 1$, we have $g_A(1) = 3 - 4\log 2$, and hence $g_\lambda(1) = 1$. By oddness, $g_\lambda(-1) = -1$. Therefore g_λ is a continuous strictly increasing map from $[-1,1]$ onto $[-1,1]$, and is consequently a bijection.

As g_λ is bijective, its inverse g_λ^{-1} exists. Using the bounds of g_λ', the inverse function theorem implies that $g_\lambda^{-1} \in C^1([-1,1])$, with $(g_\lambda^{-1})'(u) = g_\lambda'(g_\lambda^{-1}(u))^{-1}$ for all $u \in (-1,1)$, see Rudin (1976, Theorem 9.24). Using the bounds on g_λ', we obtain

$$3 - 4\log 2 \le (g_\lambda^{-1})'(u) \le K_g^{-1}.$$

By the mean value theorem, we conclude that $g_\lambda^{-1}(\cdot)$ is Lipschitz on $[-1,1]$ with the constant K_g^{-1}, which completes the proof. □

Proof of Theorem 1. We begin with a coupling result in the spirit of Bollerslev et al. (2024) and Li et al. (2024) which holds for every $1 \le i \le k_n$. To this end, let us first define the linearly combined Brownian motion for $t \in \bigcup_{i=1}^{k_n} \mathcal{I}_{n,i}$,

$$\widetilde{W}_t^{(m)} = \sigma_\tau^{(m,1)} W_t^{(1)} + \sigma_\tau^{(m,2)} W_t^{(2)}, \tag{SA.6}$$

such that $\widetilde{W}_t = (\widetilde{W}_t^{(1)}, \widetilde{W}_t^{(2)})^\top$ satisfies $\widetilde{W}_t = \sigma_\tau W_t$ for $t \in \bigcup_{i=1}^{k_n} \mathcal{I}_{n,i}$. Define the following coupled high and low returns over $\mathcal{I}_{n,i}$,

$$\widetilde{H}_{n,i}^{(m)} = \Delta_n^{-1/2}\Big(\sup_{s\in\mathcal{I}_{n,i}} \widetilde{W}_s^{(m)} - \widetilde{W}_{\tau_n+(i-1)\Delta_n}^{(m)}\Big), \quad \widetilde{L}_{n,i}^{(m)} = \Delta_n^{-1/2}\Big(\inf_{s\in\mathcal{I}_{n,i}} \widetilde{W}_s^{(m)} - \widetilde{W}_{\tau_n+(i-1)\Delta_n}^{(m)}\Big),$$

and the coupled return $\widetilde{C}_{n,i}^{(m)} = \Delta_n^{-1/2}(\widetilde{W}_{\tau_n+i\Delta_n}^{(m)} - \widetilde{W}_{\tau_n+(i-1)\Delta_n}^{(m)})$, where $\tau_n \equiv i^n(\tau)\Delta_n$ is the left endpoint of the local window $(\mathcal{I}_{n,i})_{1\le i\le k_n}$ with $\tau_n - \tau \in [0,\Delta_n), \forall n$.

We prove the following L_2-bound of the coupling error.

Lemma 3. *Under Assumption 1, it holds for all* $1 \le i \le k_n$, $Z \in \{C, H, L\}$, *and* $m \in \{1, 2\}$ *that*

$$\|\Delta_n^{-1/2} Z_{n,i}^{(m)} - \widetilde{Z}_{n,i}^{(m)}\|_2 \le K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa}. \tag{SA.7}$$

Proof. Note that the strengthened version of Assumption 1 implies the following κ-Hölder continuity in L_2-norm for all elements of σ

$$\|\sigma_t^{(m,m')} - \sigma_s^{(m,m')}\|_2 \le K|t-s|^\kappa,$$

for all $m, m' \in \{1, 2\}$ and $t, s \in [0, T]$. We now verify Equation (SA.7). Starting with the returns $Z = C$, by the triangle inequality

$$\begin{aligned}\|\Delta_n^{-1/2} C_{n,i}^{(m)} - \widetilde{C}_{n,i}^{(m)}\|_2 &\le K\Delta_n^{-1/2}\left(\left\|\int_{\mathcal{I}_{n,i}} |b_s^{(m)}|ds\right\|_2 + \sum_{\ell=1}^{2}\left\|\int_{\mathcal{I}_{n,i}} (\sigma_s^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_s^{(\ell)}\right\|_2\right)\\ &\le K\Delta_n^{1/2} + \Delta_n^{-1/2}\sum_{\ell=1}^{2}\left\|\int_{\mathcal{I}_{n,i}} (\sigma_t^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_s^{(\ell)}\right\|_2,\end{aligned}$$

where the last line follows from the boundedness of b. For the last term above, by the Itô isometry, the κ-Hölder continuity of $\sigma_t^{(m,m')}$, and the fact that $|t-\tau| \le (k_n+1)\Delta_n \le Kk_n\Delta_n$ for all $t \in \bigcup_{i=1}^{k_n}\mathcal{I}_{n,i}$, we deduce that, for $m, \ell \in \{1, 2\}$,

$$\left\|\int_{\mathcal{I}_{n,i}} (\sigma_t^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_t^{(\ell)}\right\|_2^2 = \int_{\mathcal{I}_{n,i}} \|\sigma_t^{(m,\ell)} - \sigma_\tau^{(m,\ell)}\|_2^2 dt \le K\Delta_n(k_n\Delta_n)^{2\kappa},$$

and hence $\|\Delta_n^{-1/2} C_{n,i}^{(m)} - \widetilde{C}_{n,i}^{(m)}\|_2 \le K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa}$ with $k_n \asymp \Delta_n^{-\beta}$, as desired. Now for the high return H, by repeated application of the triangle inequality,

$$\begin{aligned}|\Delta_n^{-1/2} H_{n,i}^{(m)} - \widetilde{H}_{n,i}^{(m)}| &\le \Delta_n^{-1/2}\sup_{t\in\mathcal{I}_{n,i}}\left|\int_{\tau_n+(i-1)\Delta_n}^{t} |b_s^{(m)}|ds + \sum_{\ell=1}^{2}\int_{\tau_n+(i-1)\Delta_n}^{t} (\sigma_s^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_s^{(\ell)}\right|\\ &\le \Delta_n^{-1/2}\int_{\mathcal{I}_{n,i}} |b_s^{(m)}|ds + \Delta_n^{-1/2}\sum_{\ell=1}^{2}\sup_{t\in\mathcal{I}_{n,i}}\left|\int_{\tau_n+(i-1)\Delta_n}^{t} (\sigma_s^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_s^{(\ell)}\right|\\ &\le K\Delta_n^{1/2} + \Delta_n^{-1/2}\sum_{\ell=1}^{2}\sup_{t\in\mathcal{I}_{n,i}}\left|\int_{\tau_n+(i-1)\Delta_n}^{t} (\sigma_s^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_s^{(\ell)}\right|,\end{aligned}$$

which implies that

$$\|\Delta_n^{-1/2} H_{n,i}^{(m)} - \widetilde{H}_{n,i}^{(m)}\|_2 \le K\Delta_n^{1/2} + \Delta_n^{-1/2}\sum_{\ell=1}^{2}\left\|\sup_{t\in\mathcal{I}_{n,i}}\left|\int_{\tau_n+(i-1)\Delta_n}^{t} (\sigma_s^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_s^{(\ell)}\right|\right\|_2.$$

For the last term above, by the Burkholder–Davis–Gundy inequality, we have for $m, \ell \in \{1, 2\}$

$$\left\|\sup_{t\in\mathcal{I}_{n,i}}\left|\int_{\tau_n+(i-1)\Delta_n}^{t} (\sigma_s^{(m,\ell)} - \sigma_\tau^{(m,\ell)})dW_s^{(\ell)}\right|\right\|_2^2 \le K\int_{\mathcal{I}_{n,i}} \|\sigma_t^{(m,\ell)} - \sigma_\tau^{(m,\ell)}\|_2^2 dt \le K\Delta_n(k_n\Delta_n)^{2\kappa}.$$

Therefore, we also have $\|\Delta_n^{-1/2} H_{n,i}^{(m)} - \widetilde{H}_{n,i}^{(m)}\|_2 \le K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa}$, similar to the derivation for the returns. Finally, for the low return $\|\Delta_n^{-1/2} L_{n,i}^{(m)} - \widetilde{L}_{n,i}^{(m)}\|_2$, it suffices to notice that for a

generic process X, $\inf_{t\in\mathcal{I}_{n,i}} X_t = -\sup_{t\in\mathcal{I}_{n,i}}\{-X_t\}$, and the required expectation bound follows by replacing inf with sup using essentially the same steps as in the proof for the high returns. This completes the proof. □

Write $\widetilde{A}^{(m)}_{n,i} = \widetilde{H}^{(m)}_{n,i} + \widetilde{L}^{(m)}_{n,i} - \widetilde{C}^{(m)}_{n,i}$. By the triangle inequality, one can also verify that $\|\Delta_n^{-1/2}A^{(m)}_{n,i} - \widetilde{A}^{(m)}_{n,i}\|_2 \leq K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa}$ by Lemma 3. We now identify the finite sample joint distribution of the tuple $(\widetilde{C}^{(1)}_{n,i}, \widetilde{C}^{(2)}_{n,i}, \widetilde{A}^{(1)}_{n,i}, \widetilde{A}^{(2)}_{n,i})_{1\leq i\leq k_n}$ for all n.

Lemma 4. *Conditioning on $\mathcal{F}_\tau$, it holds that*

$$(\widetilde{C}^{(1)}_{n,i}, \widetilde{C}^{(2)}_{n,i}, \widetilde{A}^{(1)}_{n,i}, \widetilde{A}^{(2)}_{n,i})_{1\leq i\leq k_n} \overset{d}{=} (\sigma^{(1)}_\tau\zeta^{(1)}_{C,i}, \sigma^{(2)}_\tau\zeta^{(2)}_{C,i}, \sigma^{(1)}_\tau\zeta^{(1)}_{A,i}, \sigma^{(2)}_\tau\zeta^{(2)}_{A,i})_{1\leq i\leq k_n}, \tag{SA.8}$$

where $(\zeta^{(1)}_{C,i}, \zeta^{(2)}_{C,i}, \zeta^{(1)}_{A,i}, \zeta^{(2)}_{A,i})_{1\leq i\leq k_n}$ are defined according to Equation (5) and the underlying Brownian motion B satisfies $\mathbb{E}[B^{(1)}_t B^{(2)}_t \mid \mathcal{F}_\tau] = \rho_\tau t, \forall t > 0$.

Proof. Recall from the definition of Σ_τ that

$$\sigma^{(m)}_\tau = \sqrt{(\sigma^{(m,1)}_\tau)^2 + (\sigma^{(m,2)}_\tau)^2}, \quad \rho_\tau = \frac{\sigma^{(1,1)}_\tau\sigma^{(2,1)}_\tau + \sigma^{(1,2)}_\tau\sigma^{(2,2)}_\tau}{\sigma^{(1)}_\tau\sigma^{(2)}_\tau}.$$

By the construction of $\widetilde{W}^{(m)}$ in Equation (SA.6), conditioning on $\mathcal{F}_\tau$ we have the following equality in law by the Markov property and the scaling property of the Brownian motion for all n,

$$\Delta_n^{-1/2}(\widetilde{W}^{(1)}_{\tau_n+s\Delta_n} - \widetilde{W}^{(1)}_{\tau_n}, \widetilde{W}^{(2)}_{\tau_n+s\Delta_n} - \widetilde{W}^{(2)}_{\tau_n})_{s\in[0,k_n]} \overset{d}{=} (\sigma^{(1)}_\tau B^{(1)}_s, \sigma^{(2)}_\tau B^{(2)}_s)_{s\in[0,k_n]},$$

which can be verified by computing the $\mathcal{F}_\tau$-conditional variance-covariance matrix for each s on both sides. Now, notice that the following two mappings are identical

$$\begin{aligned}\Delta_n^{-1/2}(\widetilde{W}^{(1)}_{\tau_n+s\Delta_n} - \widetilde{W}^{(1)}_{\tau_n}, \widetilde{W}^{(2)}_{\tau_n+s\Delta_n} - \widetilde{W}^{(2)}_{\tau_n})_{s\in[i-1,i]} &\mapsto (\widetilde{C}^{(1)}_{n,i}, \widetilde{C}^{(2)}_{n,i}, \widetilde{A}^{(1)}_{n,i}, \widetilde{A}^{(2)}_{n,i})\\ (\sigma^{(1)}_\tau B^{(1)}_s, \sigma^{(2)}_\tau B^{(2)}_s)_{s\in[i-1,i]} &\mapsto (\sigma^{(1)}_\tau\zeta^{(1)}_{C,i}, \sigma^{(2)}_\tau\zeta^{(2)}_{C,i}, \sigma^{(1)}_\tau\zeta^{(1)}_{A,i}, \sigma^{(2)}_\tau\zeta^{(2)}_{A,i}).\end{aligned}$$

Therefore, under the pushforward measure and by combining the results for all $1 \leq i \leq k_n$ exploiting independence across different i, we deduce Equation (SA.8) conditional on $\mathcal{F}_\tau$, and the proof is complete. □

We proceed to study the sample mean over k_n blocks for the original and the coupled candlestick features

$$\begin{aligned}S^{(m,m')}_n &= \frac{1}{k_n\Delta_n}\sum_{i=1}^{k_n}\big\{(1-\lambda)C^{(m)}_{n,i}C^{(m')}_{n,i} + \lambda A^{(m)}_{n,i}A^{(m')}_{n,i}\big\},\\ \widetilde{S}^{(m,m')}_n &= \frac{1}{k_n}\sum_{i=1}^{k_n}\big\{(1-\lambda)\widetilde{C}^{(m)}_{n,i}\widetilde{C}^{(m')}_{n,i} + \lambda\widetilde{A}^{(m)}_{n,i}\widetilde{A}^{(m')}_{n,i}\big\}.\end{aligned}$$

We shall also need the following existence of moments result for Brownian candlesticks, and recall the definition of $\zeta^{(m)}_{C,i}$, $\zeta^{(m)}_{H,i}$, $\zeta^{(m)}_{L,i}$ and $\zeta^{(m)}_{A,i}$ in Equations (5) and (6).

Lemma 5. *For any $k \in \mathbb{N}$, $\lambda \in [0,1]$, $m, m' \in \{1,2\}$, and $\rho \in [-1,1]$, it holds that:*

$$\mathbb{E}\big[\big|(1-\lambda)\zeta_{C,1}^{(m)}\zeta_{C,1}^{(m')} + \lambda\zeta_{A,1}^{(m)}\zeta_{A,1}^{(m')}\big|^k\big] < \infty.$$

Proof. We begin with the elementary fact that, for univariate Brownian motions $m \in \{1,2\}$, it holds that $\mathbb{E}[|\zeta_{C,1}^{(m)}|^k] = \mathbb{E}[|\zeta_{H,1}^{(m)}|^k] = \mathbb{E}[|\zeta_{L,1}^{(m)}|^k] < \infty$ for all $k \in \mathbb{N}$, where we used $|\zeta_{C,1}^{(m)}| \overset{d}{=} \zeta_{H,1}^{(m)} \overset{d}{=} -\zeta_{L,1}^{(m)}$ by the reflection principle and the reflection symmetry of Brownian motion. Therefore, by the elementary inequality $|a+b+c|^k \le K_k(|a|^k + |b|^k + |c|^k)$ for some $K_k \ge 3^{k-1}$, we have:

$$\mathbb{E}[|\zeta_{A,1}^{(m)}|^k] \le K_k(\mathbb{E}[|\zeta_{C,1}^{(m)}|^k] + \mathbb{E}[|\zeta_{H,1}^{(m)}|^k] + \mathbb{E}[|\zeta_{L,1}^{(m)}|^k]) < \infty.$$

Then, by the triangle inequality and the Cauchy–Schwarz inequality:

$$\begin{aligned}
&\mathbb{E}\big[\big|(1-\lambda)\zeta_{C,1}^{(m)}\zeta_{C,1}^{(m')} + \lambda\zeta_{A,1}^{(m)}\zeta_{A,1}^{(m')}\big|^k\big] \\
&\quad\le K_k\big(\mathbb{E}[\|\zeta_{C,1}^{(m)}\zeta_{C,1}^{(m')}\|^k] + \mathbb{E}[\|\zeta_{A,1}^{(m)}\zeta_{A,1}^{(m')}\|^k]\big) \\
&\quad\le K_k\big(\|(\zeta_{C,1}^{(m)})^k\|_2\|(\zeta_{C,1}^{(m')})^k\|_2 + \|(\zeta_{A,1}^{(m)})^k\|_2\|(\zeta_{A,1}^{(m')})^k\|_2\big) < \infty,
\end{aligned}$$

as desired. This completes the proof. □

Lemmas 3-5 lead to the following coupling relation.

Lemma 6. *For $m, m' \in \{1,2\}$ and $\lambda \in [0,1]$, it holds that*

$$\big|S_n^{(m,m')} - \widetilde{S}_n^{(m,m')}\big| = O_p(\Delta_n^{1/2\wedge(1-\beta)\kappa}). \tag{SA.9}$$

Proof. By the triangle inequality and that $\lambda \vee (1-\lambda) \le 1$,

$$\begin{aligned}
&\big|S_n^{(m,m')} - \widetilde{S}_n^{(m,m')}\big| \\
&\quad= \left|\frac{1}{k_n\Delta_n}\sum_{i=1}^{k_n}\big\{(1-\lambda)C_{n,i}^{(m)}C_{n,i}^{(m')} + \lambda A_{n,i}^{(m)}A_{n,i}^{(m')}\big\} - \frac{1}{k_n}\sum_{i=1}^{k_n}\big\{(1-\lambda)\widetilde{C}_{n,i}^{(m)}\widetilde{C}_{n,i}^{(m')} + \lambda\widetilde{A}_{n,i}^{(m)}\widetilde{A}_{n,i}^{(m')}\big\}\right| \\
&\quad\le \frac{1}{k_n}\sum_{i=1}^{k_n}\big|\Delta_n^{-1}C_{n,i}^{(m)}C_{n,i}^{(m')} - \widetilde{C}_{n,i}^{(m)}\widetilde{C}_{n,i}^{(m')}\big| + \frac{1}{k_n}\sum_{i=1}^{k_n}\big|\Delta_n^{-1}A_{n,i}^{(m)}A_{n,i}^{(m')} - \widetilde{A}_{n,i}^{(m)}\widetilde{A}_{n,i}^{(m')}\big|.
\end{aligned}$$

For the first sum above, using the triangle inequality, the identity $ab - cd = c(b-d) + d(a-c) + (a-c)(b-d)$, the Cauchy–Schwarz inequality, and Lemma 3, we have

$$\begin{aligned}
&\left\|\frac{1}{k_n}\sum_{i=1}^{k_n}\big|\Delta_n^{-1}C_{n,i}^{(m)}C_{n,i}^{(m')} - \widetilde{C}_{n,i}^{(m)}\widetilde{C}_{n,i}^{(m')}\big|\right\|_1 \\
&\quad\le \frac{1}{k_n}\sum_{i=1}^{k_n}\Big(\big\|\widetilde{C}_{n,i}^{(m)}(\Delta_n^{-1/2}C_{n,i}^{(m')} - \widetilde{C}_{n,i}^{(m')})\big\|_1 + \big\|\widetilde{C}_{n,i}^{(m')}(\Delta_n^{-1/2}C_{n,i}^{(m)} - \widetilde{C}_{n,i}^{(m)})\big\|_1 \\
&\qquad\quad + \big\|(\Delta_n^{-1/2}C_{n,i}^{(m')} - \widetilde{C}_{n,i}^{(m')})(\Delta_n^{-1/2}C_{n,i}^{(m)} - \widetilde{C}_{n,i}^{(m)})\big\|_1\Big) \\
&\quad\le (K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa})(1 + K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa}) \\
&\quad\le K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa}.
\end{aligned} \tag{SA.10}$$

where the penultimate line follows from the fact that $\|\widetilde{C}_{n,i}^{(m)}\|_2^2 = \|\sigma_\tau^{(m)}\|_2^2 \leq K$ by the boundedness of $\sigma^{(m)}$. Using exactly the same decomposition and that $\|\widetilde{A}_{n,i}^{(m)}\|_2^2 \leq K$ again by the boundedness of $\sigma^{(m)}$ and Lemmas 4 and 5, we conclude that

$$\left\| \frac{1}{k_n} \sum_{i=1}^{k_n} \left| \Delta_n^{-1} A_{n,i}^{(m)} A_{n,i}^{(m')} - \widetilde{A}_{n,i}^{(m)} \widetilde{A}_{n,i}^{(m')} \right| \right\|_1 \leq K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge(1-\beta)\kappa}. \tag{SA.11}$$

Combining Equations (SA.10) and (SA.11), we deduce (SA.9) by Markov's inequality, and the proof is complete. □

Lemma 6 suggests that $\left|S_n^{(m,m')} - \widetilde{S}_n^{(m,m')}\right| = o_p(1)$ for all $\beta < 1$, and if $\beta < 2\kappa/(2\kappa+1)$, then we further have $\sqrt{k_n}\left|S_n^{(m,m')} - \widetilde{S}_n^{(m,m')}\right| = o_p(1)$. As a result, we only need to prove the convergence in probability and the convergence in distribution for $\widetilde{S}_n^{(m,m')}$, as the same convergence also holds for $S_n^{(m,m')}$ by Lemma 1.

We proceed to study the asymptotic behavior of $\widetilde{S}_n^{(m,m')}$ as $k_n \to \infty$. We begin by identifying the limit of $\widetilde{S}_n^{(m,m')}$.

Lemma 7. *Let $\theta_\lambda = 1 + 2\lambda(1 - 2\log 2)$. It holds as $k_n \to \infty$ that*

$$\widetilde{S}_n^{(m,m)} \xrightarrow{\mathbb{P}} \theta_\lambda (\sigma_\tau^{(m)})^2, \quad \widetilde{S}_n^{(1,2)} \xrightarrow{\mathbb{P}} \theta_\lambda \sigma_\tau^{(1)} \sigma_\tau^{(2)} g_\lambda(\rho_\tau).$$

Proof. We shall only prove the convergence for $\widetilde{S}_n^{(1,2)}$, as the same proof applies to $\widetilde{S}_n^{(m,m)}$ in the case of perfect correlation, i.e., $g_\lambda(1) = 1$. We begin with the following fact, which holds for all i and n,

$$\mathbb{E}[(1-\lambda)\widetilde{C}_{n,i}^{(1)}\widetilde{C}_{n,i}^{(2)} + \lambda \widetilde{A}_{n,i}^{(1)}\widetilde{A}_{n,i}^{(2)} \mid \mathcal{F}_{\tau_n+(i-1)\Delta_n}] = \theta_\lambda \sigma_\tau^{(1)}\sigma_\tau^{(2)} g_\lambda(\rho_\tau).$$

This follows directly from Lemma 4 and the Brownian moment identity in (8):

$$\mathbb{E}[(1-\lambda)\zeta_{C,1}^{(1)}\zeta_{C,1}^{(2)} + \lambda \zeta_{A,1}^{(1)}\zeta_{A,1}^{(2)}] = \theta_\lambda g_\lambda(\rho). \tag{SA.12}$$

Also, by Lemmas 4 and 5, we deduce

$$\mathbb{E}\big[\big((1-\lambda)\widetilde{C}_{n,i}^{(1)}\widetilde{C}_{n,i}^{(2)} + \lambda \widetilde{A}_{n,i}^{(1)}\widetilde{A}_{n,i}^{(2)}\big)^2 \mid \mathcal{F}_{\tau_n+(i-1)\Delta_n}\big] < \infty,$$

and hence

$$\frac{1}{k_n^2}\sum_{i=1}^{k_n} \mathbb{E}\big[\big((1-\lambda)\widetilde{C}_{n,i}^{(1)}\widetilde{C}_{n,i}^{(2)} + \lambda \widetilde{A}_{n,i}^{(1)}\widetilde{A}_{n,i}^{(2)}\big)^2 \mid \mathcal{F}_{\tau_n+(i-1)\Delta_n}\big] \to 0.$$

Then, by Jacod and Protter (2012, Lemma 2.2.11), the triangular array $(z_i^n)_{1\leq i\leq k_n}$ defined by

$$z_i^n = \frac{1}{k_n}\big((1-\lambda)\widetilde{C}_{n,i}^{(1)}\widetilde{C}_{n,i}^{(2)} + \lambda \widetilde{A}_{n,i}^{(1)}\widetilde{A}_{n,i}^{(2)} - \theta_\lambda \sigma_\tau^{(1)}\sigma_\tau^{(2)} g_\lambda(\rho_\tau)\big),$$

is asymptotically negligible, which implies that $\sum_{i=1}^{k_n} z_i^n \xrightarrow{\mathbb{P}} 0$. This completes the proof. □

By Lemma 2, $g_\lambda^{-1}(\cdot)$ is continuous on $[-1,1]$. Therefore, Lemma 7 and the continuous mapping theorem lead to

$$\tilde{\rho}_n(\lambda) \equiv g_\lambda^{-1}\left(\frac{\widetilde{S}_n^{(1,2)}}{\sqrt{\widetilde{S}_n^{(1,1)}\widetilde{S}_n^{(2,2)}}}\right) \xrightarrow{\mathbb{P}} \rho_\tau. \tag{SA.13}$$

Also, using the Lipschitz property of $g_\lambda^{-1}(\cdot)$ on $[-1,1]$ by Lemma 2, Lemmas 6 and 7 give the following estimate

$$\begin{aligned}
|\hat{\rho}_n(\lambda) - \tilde{\rho}_n(\lambda)| &= \left| g_\lambda^{-1}\left(\frac{S_n^{(1,2)}}{\sqrt{S_n^{(1,1)}S_n^{(2,2)}}}\right) - \tilde{\rho}_n(\lambda) \right| \\
&\leq K\left| \frac{\widetilde{S}_n^{(1,2)} + o_p(1)}{\sqrt{\widetilde{S}_n^{(1,1)}\widetilde{S}_n^{(2,2)}} + o_p(1)} - \frac{\widetilde{S}_n^{(1,2)}}{\sqrt{\widetilde{S}_n^{(1,1)}\widetilde{S}_n^{(2,2)}}} \right| \\
&\leq K\left| \frac{(\sqrt{\widetilde{S}_n^{(1,1)}\widetilde{S}_n^{(2,2)}} - \widetilde{S}_n^{(1,2)}) \cdot o_p(1)}{\widetilde{S}_n^{(1,1)}\widetilde{S}_n^{(2,2)} + \sqrt{\widetilde{S}_n^{(1,1)}\widetilde{S}_n^{(2,2)}} \cdot o_p(1)} \right| = o_p(1).
\end{aligned} \tag{SA.14}$$

Combining Equations (SA.13) with (SA.14) yields (10).

Moving on to the proof of (11). From Lemma 6 and Lemma 7 and under the additional condition $\beta < 2\kappa/(2\kappa+1)$, we can replace all the $o_p(1)$ terms by $o_p(k_n^{-1/2})$ which improves the estimate in (SA.14) to $\sqrt{k_n}|\hat{\rho}_n(\lambda) - \tilde{\rho}_n(\lambda)| = o_p(1)$. Therefore, it suffices to prove the same $\mathcal{F}_\tau$-conditional convergence for $\tilde{\rho}_n(\lambda)$ to conclude (11) using the asymptotic equivalence lemma. To prove this, consider the following vector-valued triangular array $\eta_{n,i} = (\eta_{n,i}^{(1,2)}, \eta_{n,i}^{(1,1)}, \eta_{n,i}^{(2,2)})^\top$, where

$$\begin{aligned}
\eta_{n,i}^{(1,2)} &= \frac{1}{\sqrt{k_n}}\left(\frac{(1-\lambda)\widetilde{C}_{n,i}^{(1)}\widetilde{C}_{n,i}^{(2)} + \lambda\widetilde{A}_{n,i}^{(1)}\widetilde{A}_{n,i}^{(2)}}{\theta_\lambda \sigma_\tau^{(1)}\sigma_\tau^{(2)}} - g_\lambda(\rho_\tau)\right), \\
\eta_{n,i}^{(m,m)} &= \frac{1}{\sqrt{k_n}}\left(\frac{(1-\lambda)\widetilde{C}_{n,i}^{(m)}\widetilde{C}_{n,i}^{(m)} + \lambda\widetilde{A}_{n,i}^{(m)}\widetilde{A}_{n,i}^{(m)}}{\theta_\lambda (\sigma_\tau^{(m)})^2} - 1\right), \quad m \in \{1,2\},
\end{aligned}$$

which is, by Equation (SA.12), a vector-valued martingale difference sequence for each n with all moments finite. By Lemma 4, we see that

$$\mathbb{E}[\eta_{n,i}\eta_{n,i}^\top \mid \mathcal{F}_{\tau_n+(i-1)\Delta_n}] = \frac{1}{k_n}\Sigma_\lambda(\rho_\tau), \quad \forall 1 \leq i \leq k_n,$$

where $\Sigma_\lambda(\rho) = [\Sigma_\lambda^{(m,m')}(\rho)]_{m,m'\in\{1,2,3\}}$ is a 3-by-3 symmetric matrix with the following entries:

$$\begin{aligned}
&\Sigma_\lambda^{(1,1)}(\rho) = \mathrm{Var}(\zeta_\lambda^{(1,2)}), \quad \Sigma_\lambda^{(1,2)}(\rho) = \Sigma_\lambda^{(1,3)}(\rho) = \mathrm{Cov}(\zeta_\lambda^{(1,2)}, \zeta_\lambda^{(1,1)}), \\
&\Sigma_\lambda^{(2,2)}(\rho) = \Sigma_\lambda^{(3,3)}(\rho) = \mathrm{Var}(\zeta_\lambda^{(1,1)}), \quad \Sigma_\lambda^{(2,3)}(\rho) = \mathrm{Cov}(\zeta_\lambda^{(1,1)}, \zeta_\lambda^{(2,2)}),
\end{aligned}$$

in which $\zeta_\lambda^{(m,m')} = [(1-\lambda)\zeta_{C,1}^{(m)}\zeta_{C,1}^{(m')} + \lambda\zeta_{A,1}^{(m)}\zeta_{A,1}^{(m')}]/\theta_\lambda$. Also, for any fixed $\varepsilon > 0$ and $a \in \mathbb{R}^3$,

$$\begin{aligned}
&\sum_{i=1}^{k_n} \mathbb{E}[(a^\top\eta_{n,i})^2 \mathbb{1}_{\{|a^\top\eta_{n,i}|>\varepsilon\}} \mid \mathcal{F}_{\tau_n+(i-1)\Delta_n}] \leq \frac{1}{\varepsilon^2}\sum_{i=1}^{k_n} \mathbb{E}[(a^\top\eta_{n,i})^4 \mid \mathcal{F}_{\tau_n+(i-1)\Delta_n}] \\
&\quad \leq \frac{K\|a\|_\infty^4}{\varepsilon^2}\sum_{i=1}^{k_n} k_n^{-2}\mathbb{E}[(\zeta_\lambda^{(1,1)})^4 + (\zeta_\lambda^{(2,2)})^4 + (\zeta_\lambda^{(1,2)})^4] \xrightarrow{\mathbb{P}} 0,
\end{aligned} \tag{SA.15}$$

where $\|a\|_\infty$ is the sup norm of a, and the last convergence follows by Lemma 5. Therefore, by Jacod and Protter (2012, Lemma 2.2.11) we deduce:

$$\sum_{i=1}^{k_n} \mathbb{E}[\eta_{n,i}\eta_{n,i}^\top \mid \mathcal{F}_{\tau_n+(i-1)\Delta_n}] \xrightarrow{\mathbb{P}} \Sigma_\lambda(\rho_\tau) \in \mathcal{F}_\tau. \tag{SA.16}$$

By the Cramér–Wold device and using the corollary of Theorem 3 of Eagleson (1975) with the two conditions in Equations (SA.15) and (SA.16) satisfied, we conclude that:

$$\sqrt{k_n}\left[\begin{pmatrix} (\theta_\lambda \sigma_\tau^{(1)}\sigma_\tau^{(2)})^{-1}\widetilde{S}_n^{(1,2)} \\ \left(\theta_\lambda (\sigma_\tau^{(1)})^2\right)^{-1}\widetilde{S}_n^{(1,1)} \\ \left(\theta_\lambda (\sigma_\tau^{(2)})^2\right)^{-1}\widetilde{S}_n^{(2,2)} \end{pmatrix} - \begin{pmatrix} g_\lambda(\rho_\tau) \\ 1 \\ 1 \end{pmatrix}\right] \equiv \sum_{i=1}^{k_n} \eta_{n,i} \xrightarrow{\mathcal{L}|\mathcal{F}_\tau} \Sigma_\lambda(\rho_\tau)^{1/2} Z_3, \tag{SA.17}$$

where Z_3 is a 3-dimensional standard Gaussian variable defined on an extension of $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ independent of $\mathcal{F}$, and $\Sigma_\lambda(\rho_\tau)^{1/2}$ is the (unique) square-root matrix of $\Sigma_\lambda(\rho_\tau)$.

Now, by Lemma 2, $u \mapsto g_\lambda^{-1}(u)$ is continuously differentiable on $(-1,1)$ for all $\lambda \in [0,1]$ with derivative bounded and bounded away from zero. Since ρ_τ is bounded away from ± 1 by the strengthened version of Assumption 1, $g_\lambda(\rho_\tau)$ is bounded away from ± 1, and hence g_λ^{-1} is differentiable at $g_\lambda(\rho_\tau)$ with $0 < (g_\lambda^{-1})'(g_\lambda(\rho_\tau)) = g_\lambda'(\rho_\tau)^{-1} < \infty$. An application of the multivariate delta method to (SA.17) under the mapping $(x,y,z)^\top \mapsto g_\lambda^{-1}(x/\sqrt{yz})$ yields:

$$\sqrt{k_n}(\tilde{\rho}_n(\lambda) - \rho_\tau) \equiv \sqrt{k_n}\left[g_\lambda^{-1}\left(\frac{\widetilde{S}_n^{(1,2)}}{\sqrt{\widetilde{S}_n^{(1,1)}\widetilde{S}_n^{(2,2)}}}\right) - \rho_\tau\right] \xrightarrow{\mathcal{L}|\mathcal{F}_\tau} g_\lambda'(\rho_\tau)^{-1}\sqrt{V_\lambda(\rho_\tau)}Z, \tag{SA.18}$$

where, with $\Xi_\lambda(\rho_\tau) \equiv \zeta_\lambda^{(1,2)} - g_\lambda(\rho_\tau)(\zeta_\lambda^{(1,1)} + \zeta_\lambda^{(2,2)})/2$ and noting that $\mathbb{E}[\Xi_\lambda(\rho_\tau)] = 0$,

$$\begin{aligned} V_\lambda(\rho_\tau) &= \mathrm{Var}\big(\Xi_\lambda(\rho_\tau) \mid \mathcal{F}_\tau\big) = \mathbb{E}[\Xi_\lambda(\rho_\tau)^2 \mid \mathcal{F}_\tau] \\ &= \Sigma_\lambda^{(1,1)}(\rho_\tau) + \frac{g_\lambda(\rho_\tau)^2}{2}\big(\Sigma_\lambda^{(2,2)}(\rho_\tau) + \Sigma_\lambda^{(2,3)}(\rho_\tau)\big) - 2g_\lambda(\rho_\tau)\Sigma_\lambda^{(1,2)}(\rho_\tau). \end{aligned}$$

We note that $V_\lambda(\rho_\tau) > 0$ for all $\rho_\tau \in (-1,1)$, so the limit in Equation (SA.18) is non-degenerate. To see this, it suffices to prove that $\Xi_\lambda(\rho_\tau)$ is not almost surely 0 for $\rho_\tau \in (-1,1)$. Indeed, Lemma 2 gives $g_\lambda(\rho) \in (-1,1)$. Now pick any deterministic continuous paths h that generate $\zeta_{C,1}^{(m)} = \zeta_{A,1}^{(m)} = 1$ for $m \in \{1,2\}$, e.g., a linear interpolation of $h(0) = 0$, $h(t) = 2$ and $h(1) = 1$ for $t \in (0,1)$. On the bivariate path (h,h), we have $\zeta_\lambda^{(m,m')} \equiv \theta_\lambda^{-1}$ and hence $\Xi_\lambda(\rho_\tau) = (1 - g_\lambda(\rho_\tau))/\theta_\lambda > 0$. By continuity of the candlestick functionals, there exists an open neighbourhood of (h,h) such that $\Xi_\lambda(\rho_\tau) > 0$. Since $(B^{(1)}, B^{(2)})$ has full support on $C_0([0,1])^2$, such an open neighbourhood has positive probability measure, and hence $\Xi_\lambda(\rho_\tau)$ is not almost surely zero. It is also easy to verify that $V_\lambda(\pm 1) = 0$, which is prohibited by the bound of ρ_τ in Assumption 1.

Combining Equation (SA.18) and the estimate $\sqrt{k_n}|\hat{\rho}_n(\lambda) - \tilde{\rho}_n(\lambda)| = o_p(1)$ yields (11) by Lemma 1(ii), and the proof is complete. □

Proof of Corollary 1. By the $\mathcal{F}_\tau$-conditional CLT in Theorem 1, the feasible plug-in CLT follows from the continuous mapping theorem and Slutsky's theorem provided that, on $(-1,1)$: (1) g'_λ and V_λ are positive, and (2) g'_λ and V_λ are continuous in ρ. Lemma 2 proves that $g'_\lambda > 0$ and the continuity of g'_λ, and $V_\lambda > 0$ is proved in the proof of Theorem 1.

It remains to prove that V_λ is continuous in ρ. Following the same technique for the continuity of g_λ in the proof of Lemma 2, we take any sequence $\rho_n \to \rho \in (-1,1)$. Since $\Xi_\lambda(\rho)^2$ is a continuous function of the candlestick functionals, which are in turn continuous in the Brownian paths that are continuous in ρ almost surely, we have $\Xi_\lambda(\rho_n)^2 \to \Xi_\lambda(\rho)^2$ almost surely as $\rho_n \to \rho$. Furthermore, recall $M \equiv \sup_{0\le t\le 1}|B_t| + \sup_{0\le t\le 1}|B_t^\perp|$. For $m, m' \in \{1,2\}$ and M constructed from the corresponding driving Brownian motions, we have the following bound independent of ρ and $\lambda \in [0,1]$:

$$|\zeta_\lambda^{(m,m')}| \le \frac{(1-\lambda)|\zeta_{C,1}^{(m)}\zeta_{C,1}^{(m')}| + \lambda|\zeta_{A,1}^{(m)}\zeta_{A,1}^{(m')}|}{\theta_\lambda} \le \frac{1+8\lambda}{\theta_\lambda}M^2 \le KM^2,$$

where the last estimate uses $(1+8\lambda)/\theta_\lambda \le 9/(3-4\log 2)$ for $\lambda \in [0,1]$. Therefore,

$$\Xi_\lambda(\rho)^2 \le \left(|\zeta_\lambda^{(1,2)}| + \frac{|g_\lambda(\rho)|}{2}(|\zeta_\lambda^{(1,1)}| + |\zeta_\lambda^{(2,2)}|)\right)^2 \le KM^4.$$

Recall from the proof of Lemma 2 that $\mathbb{E}[M^4] < \infty$. Applying the dominated convergence theorem and using the continuity of g_λ, we conclude that

$$V_\lambda(\rho_n) = \mathbb{E}[\Xi_\lambda(\rho_n)^2] \to \mathbb{E}[\Xi_\lambda(\rho)^2] = V_\lambda(\rho),$$

proving the continuity of V_λ on $(-1,1)$. This completes the proof. □

Proof of Theorem 2. Recall that under the fixed-k setting, we have $\beta = 0$. Therefore, Lemma 3 gives

$$\|\Delta_n^{-1/2}C_{n,i}^{(m)} - \widetilde{C}_{n,i}^{(m)}\|_2 \le K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge\kappa},$$
$$\|\Delta_n^{-1/2}A_{n,i}^{(m)} - \widetilde{A}_{n,i}^{(m)}\|_2 \le K\Delta_n^{1/2} + K'\Delta_n^{1/2\wedge\kappa},$$

which implies that, jointly for all $1 \le i \le k$,

$$\Delta_n^{-1/2}(C_{n,i}^{(1)}, C_{n,i}^{(2)}, A_{n,i}^{(1)}, A_{n,i}^{(2)}) = (\widetilde{C}_{n,i}^{(1)}, \widetilde{C}_{n,i}^{(2)}, \widetilde{A}_{n,i}^{(1)}, \widetilde{A}_{n,i}^{(2)}) + o_p(1).$$

By Lemma 1(i), for all $\lambda \in [0,1]$, we further have

$$\hat{\rho}_n(\lambda) = \tilde{\rho}_n(\lambda) + o_p(1), \tag{SA.19}$$

and recall $\tilde{\rho}_n(\lambda)$ defined in Equation (SA.13).

It remains to notice that, by Lemma 4 and conditioning on $\mathcal{F}_\tau$, we have the following equality in distribution conditional on $\mathcal{F}_\tau$ independent of n

$$\tilde{\rho}_n(\lambda) \overset{\mathcal{L}|\mathcal{F}_\tau}{=} \tilde{\rho}(\lambda;\rho_\tau).$$

The above result directly leads to the desired $\mathcal{F}_\tau$-conditional convergence in law in Equation (17) by (SA.19) and the asymptotic equivalence lemma. This completes the proof. □

Proof of Corollary 2. By the assumed $\mathcal{F}_\tau$-conditional non-atomicity of $\tilde{\rho}(\lambda;\rho_\tau)$, its boundary has zero conditional probability

$$\mathbb{P}\big(\tilde{\rho}(\lambda;\rho_\tau) \in \partial I_{1-\alpha}(\lambda;\rho_\tau) \mid \mathcal{F}_\tau\big) = 0, \quad \text{a.s.}$$

Hence $I_{1-\alpha}(\lambda;\rho_\tau)$ is a conditional continuity set for the limiting law. Therefore, Theorem 2 and the conditional portmanteau theorem give

$$\mathbb{P}\big(\hat{\rho}_n(\lambda) \in I_{1-\alpha}(\lambda;\rho_\tau) \mid \mathcal{F}_\tau\big) \xrightarrow{\mathbb{P}} \mathbb{P}\big(\tilde{\rho}(\lambda;\rho_\tau) \in I_{1-\alpha}(\lambda;\rho_\tau) \mid \mathcal{F}_\tau\big) = 1-\alpha,$$

where the equality follows from the construction of $I_\lambda(\lambda;\rho_\tau)$ and the non-atomic condition. Now by the definition of the inverted confidence set

$$\{\rho_\tau \in \mathrm{CS}_{n,1-\alpha}(\lambda)\} = \{\hat{\rho}_n(\lambda) \in I_{1-\alpha}(\lambda;\rho_\tau)\}.$$

Therefore,

$$\mathbb{P}\big(\rho_\tau \in \mathrm{CS}_{n,1-\alpha}(\lambda) \mid \mathcal{F}_\tau\big) = \mathbb{P}\big(\hat{\rho}_n(\lambda) \in I_{1-\alpha}(\lambda;\rho_\tau) \mid \mathcal{F}_\tau\big) \xrightarrow{\mathbb{P}} 1-\alpha,$$

which completes the proof.

□

SB (Near) Minimaxity in the Quadratic Consistent Class

Our numerical results in Figure 2 of the main text indicate that $\widetilde{V}(\lambda;\rho)$ is nonincreasing in $|\rho|$ for every $\lambda \in [0,1]$, and in particular at λ^*. A more granular numerical illustration is presented in Figure S1. Unfortunately, verifying this monotonicity condition requires new analytical results for the fourth moments of correlated Brownian motions, which are not yet available in the literature. However, under the weaker condition that $\widetilde{V}(\lambda;0) \geq \widetilde{V}(\lambda;\rho)$ for all $\rho \in [-1,1]$, we shall show that the claimed minimaxity result in the main text actually holds over a much wider class than the current one-parameter class of estimators $\mathcal{E}_{\mathrm{L}} = \{\hat{\rho}_n(\lambda) : \lambda \in [0,1]\}$.

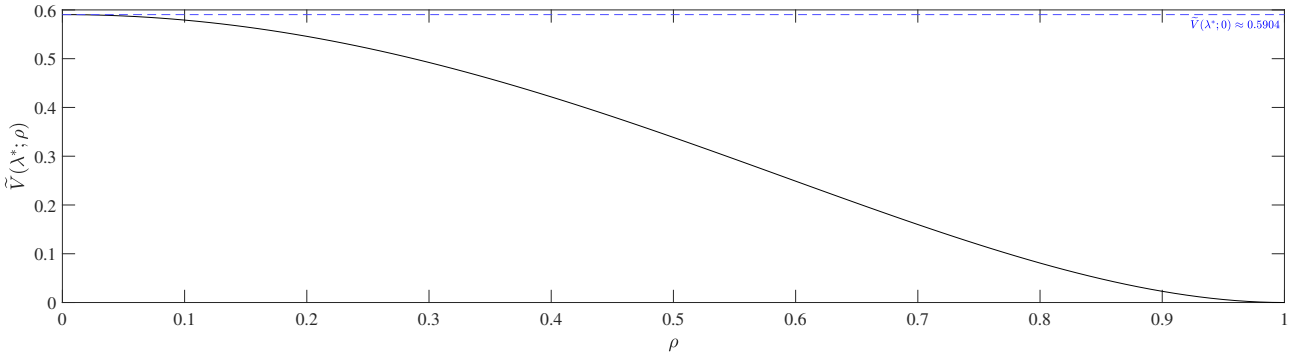


Figure S1: Numerically computed $\widetilde{V}(\lambda^*;\rho)$. The moments in $V_{\lambda^*}(\rho)$ are simulated from 10^7 realizations of bivariate Brownian paths with 10^6 Euler discretization steps.

To fix ideas, let us define the vector of averaged quadratic normalized candlestick features

$$X_n = \frac{1}{k_n \Delta_n} \sum_{i=1}^{k_n} \big(C_{n,i}^{(1)} C_{n,i}^{(2)}, A_{n,i}^{(1)} A_{n,i}^{(2)}, (C_{n,i}^{(1)})^2, (A_{n,i}^{(1)})^2, (C_{n,i}^{(2)})^2, (A_{n,i}^{(2)})^2\big)^\top,$$

associated with the moment vector

$$m(\Sigma) \equiv \big(\sigma^{(1)}\sigma^{(2)}\rho, \sigma^{(1)}\sigma^{(2)} g_A(\rho), (\sigma^{(1)})^2, (\sigma^{(1)})^2\theta_1, (\sigma^{(2)})^2, (\sigma^{(2)})^2\theta_1\big)^\top,$$

where Σ is the 2-by-2 variance-covariance matrix with volatilities $\sigma^{(1)}$, $\sigma^{(2)}$ and correlation ρ, $g_A(\rho) = 2(c(\rho) - c(-\rho)) - \rho$, and $\theta_\lambda = 1 + 2\lambda(1 - 2\log 2)$. Note that $m(\Sigma) = \mathbb{E}[\zeta_{X,1}(\Sigma)]$ with

$$\zeta_{X,1}(\Sigma) \equiv \big(\sigma^{(1)}\sigma^{(2)}\zeta_{C,1}^{(1)}\zeta_{C,1}^{(2)}, \sigma^{(1)}\sigma^{(2)}\zeta_{A,1}^{(1)}\zeta_{A,1}^{(2)}, (\sigma^{(1)}\zeta_{C,1}^{(1)})^2, (\sigma^{(1)}\zeta_{A,1}^{(1)})^2, (\sigma^{(2)}\zeta_{C,1}^{(2)})^2, (\sigma^{(2)}\zeta_{A,1}^{(2)})^2\big)^\top,$$

where $(\zeta_{C,1}^{(1)}\zeta_{C,1}^{(2)}, \zeta_{A,1}^{(1)}\zeta_{A,1}^{(2)})$ are constructed from a bivariate Brownian motion with correlation ρ.

For any function $h : \mathbb{R}^6 \to [-1, 1]$, define a candidate spot correlation estimator as:

$$\hat{\rho}_n(h) = h(X_n).$$

For $\hat{\rho}_n(h)$ to be a well-behaved correlation estimator, we impose some regularity conditions on h.

Define the open domain in $\mathbb{R}^6$:

$$U = \{x \in \mathbb{R}^6 : x_3 \wedge x_4 \wedge x_5 \wedge x_6 > 0, x_1^2 < x_3 x_5, x_2^2 < x_4 x_6\},$$

and let $\mathcal{M} := \{m(\Sigma) : \Sigma \in S_{++}^2\} \subset U$, where S_{++}^2 is the 2-by-2 positive-definite cone. We define the class $\mathcal{H}_{\mathrm{QC}}$ as the set of functions $h : U \to [-1, 1]$ such that:

(i) h is continuously differentiable on U;

(ii) h satisfies the consistency condition

$$h\big(m(\Sigma)\big) = \rho, \qquad \forall \Sigma \in S_{++}^2,$$

where ρ is the associated correlation coefficient of Σ;

(iii) h is scale-invariant:

$$h(q_1 q_2 x_1, q_1 q_2 x_2, q_1^2 x_3, q_1^2 x_4, q_2^2 x_5, q_2^2 x_6) = h(x)$$

for all $q_1, q_2 > 0$ and all $x \in U$.

In particular, writing $\Sigma = D^{1/2} R_\rho D^{1/2}$, where D is the diagonal matrix of variances and R_ρ is the associated correlation matrix, then the scale-invariance and consistency conditions of h imply $h(m(\Sigma)) = h(m(R_\rho)) = \rho$.

We define the following quadratic consistent class of correlation estimators as

$$\mathcal{E}_{\mathrm{QC}} \equiv \{\hat{\rho}_n(h) : h \in \mathcal{H}_{\mathrm{QC}}\}.$$

The class $\mathcal{E}_{\mathrm{QC}}$ clearly nests the linear class of estimators with $\mathcal{E}_{\mathrm{L}} \subset \mathcal{E}_{\mathrm{QC}}$. It also includes any (possibly infinite) convex combinations of elements in $\mathcal{E}_{\mathrm{L}}$, such as the Popov estimator $\hat{\rho}_{P,n} = (\hat{\rho}_n(0) + \hat{\rho}_n(1))/2$, as well as consistent modifications of linear combinations of the inconsistent spot correlation estimators in $\{\hat{\varrho}_n(\lambda) : \lambda \in [0,1]\}$. Consequently, it covers a substantially larger class of estimators than $\mathcal{E}_{\mathrm{L}}$ studied in the main text.

By Theorem 1 and the multivariate delta method, it is straightforward to show that all elements in $\mathcal{E}_{\mathrm{QC}}$ are consistent and asymptotically $\mathcal{F}_\tau$-conditionally Gaussian. To present the limit law, we introduce some further notation. Denote $m_\tau \equiv m(\Sigma_\tau)$, and write $\nabla h(m_\tau)$ as the 6-by-1 gradient vector of h evaluated at m_τ. Define the following conditional variance-covariance matrix:

$$\mathrm{Var}\big(\zeta_{X,1}(\Sigma_\tau) \mid \mathcal{F}_\tau\big) = \Omega_\tau,$$

where R_ρ is the 2-by-2 correlation matrix with correlation coefficient ρ, and write $\widetilde{V}_{\mathrm{QC}}(h;\rho_\tau) \equiv \nabla h(m_\tau)^\top \Omega_\tau \nabla h(m_\tau)$. Note that $\widetilde{V}_{\mathrm{QC}}(h;\rho_\tau)$ depends on Σ_τ only through ρ_τ due to the scale-invariance of h. We have the following result:

Corollary S1. *Under the same conditions as the consistency result of Theorem 1, it holds for all fixed $h \in \mathcal{H}_{QC}$ that*

$$\hat{\rho}_n(h) \xrightarrow{\mathbb{P}} \rho_\tau.$$

Furthermore, under the same condition as the $\mathcal{F}_\tau$-conditional CLT of Theorem 1:

$$\sqrt{k_n}\big(\hat{\rho}_n(h) - \rho_\tau\big) \xrightarrow{\mathcal{L}|\mathcal{F}_\tau} \sqrt{\widetilde{V}_{\mathrm{QC}}(h;\rho_\tau)}Z,$$

where $Z \sim \mathcal{N}(0,1)$ is defined on an extension of $(\Omega, (\mathcal{F}_t)_{t\geq 0}, \mathcal{F}, \mathbb{P})$ independent of $\mathcal{F}$.

We now show that, under the numerically verified inequality $\widetilde{V}(\lambda;0) \geq \widetilde{V}(\lambda;\rho)$, our proposed estimator $\hat{\rho}_n^*$ is in fact minimax in the class $\mathcal{E}_{\mathrm{QC}}$.

Theorem S1. *Suppose $\widetilde{V}(\lambda^*;\rho) \leq \widetilde{V}(\lambda^*;0), \forall \rho \in (-1,1)$. Then*

$$\inf_{h\in\mathcal{H}_{\mathrm{QC}}} \sup_{\rho\in(-1,1)} \widetilde{V}_{\mathrm{QC}}(h;\rho) = \sup_{\rho\in(-1,1)} \widetilde{V}(\lambda^*;\rho) = \widetilde{V}(\lambda^*;0).$$

Thus $\hat{\rho}_n^* = \hat{\rho}_n(\lambda^*)$ is asymptotically minimax within the quadratic consistent class $\mathcal{E}_{\mathrm{QC}}$.

SB.1 Proofs of Corollary S1 and Theorem S1

Proof of Corollary S1. Define the following coupled version of X_n:

$$\widetilde{X}_n \equiv \frac{1}{k_n}\sum_{i=1}^{k_n}\big(\widetilde{C}_{n,i}^{(1)}\widetilde{C}_{n,i}^{(2)}, \widetilde{A}_{n,i}^{(1)}\widetilde{A}_{n,i}^{(2)}, (\widetilde{C}_{n,i}^{(1)})^2, (\widetilde{A}_{n,i}^{(1)})^2, (\widetilde{C}_{n,i}^{(2)})^2, (\widetilde{A}_{n,i}^{(2)})^2\big)^\top.$$

By the same proof as Lemma 6, we have the following coupling relation

$$\|X_n - \widetilde{X}_n\|_1 = O_p(\Delta_n^{1/2\wedge(1-\beta)\kappa}). \tag{SB.20}$$

Furthermore, by the same proof as Lemma 7, for $\beta \in (0,1)$ we deduce $\widetilde{X}_n \xrightarrow{\mathbb{P}} m(\Sigma_\tau)$. The coupling relation in (SB.20) and the continuous mapping lemma allow us to pass the convergence to the observed candlestick feature vector: $X_n \xrightarrow{\mathbb{P}} m(\Sigma_\tau)$. Finally, for any $h \in \mathcal{H}_{\mathrm{QC}}$, by the continuity of h at $m(\Sigma_\tau)$ and the consistency condition of h, we deduce:

$$\hat{\rho}_n(h) = h(X_n) \xrightarrow{\mathbb{P}} h\big(m(\Sigma_\tau)\big) = \rho_\tau, \tag{SB.21}$$

which proves the consistency of $\hat{\rho}_n(h) \in \mathcal{E}_{\mathrm{QC}}$.

For the conditional CLT, following similar steps as the proof of (SA.17), with $\beta < 2\kappa/(2\kappa+1)$ we also deduce the following multivariate $\mathcal{F}_\tau$-conditional CLT:

$$\sqrt{k_n}\big(\widetilde{X}_n - m(\Sigma_\tau)\big) \xrightarrow{\mathcal{L}|\mathcal{F}_\tau} Z_\tau,$$

where Z_τ is a centered conditionally Gaussian vector with the $\mathcal{F}_\tau$-conditional covariance matrix $\mathrm{Var}(\zeta_{X,1}(\Sigma_\tau) \mid \mathcal{F}_\tau)$. By (SB.20) and the asymptotic equivalence lemma, we can also pass this hexvariate CLT to the observed candlestick feature vector:

$$\sqrt{k_n}\big(X_n - m(\Sigma_\tau)\big) \xrightarrow{\mathcal{L}|\mathcal{F}_\tau} Z_\tau. \tag{SB.22}$$

Now take any $h \in \mathcal{H}_{\mathrm{QC}}$. Since h is continuously differentiable at $m(\Sigma_\tau)$ by assumption, applying the multivariate delta method to the above conditional CLT yields

$$\sqrt{k_n}\big(\hat{\rho}_n(h) - \rho_\tau\big) = \sqrt{k_n}\big(h(X_n) - h(m(\Sigma_\tau))\big) \xrightarrow{\mathcal{L}|\mathcal{F}_\tau} \nabla h(m_\tau)^\top Z_\tau,$$

and it suffices to notice that $\nabla h(m_\tau)^\top Z_\tau$ and $\sqrt{\widetilde{V}_{\mathrm{QC}}(h;\rho_\tau)}Z$ are equal in $\mathcal{F}_\tau$-conditional law by the scale invariance of h. This completes the proof. □

Proof of Theorem S1. We first establish a lower bound for the risk at $\rho = 0$. Let R_ρ denote the 2-by-2 correlation matrix identified by $\rho \in (-1,1)$, and hence R_0 is the 2-dimensional identity matrix. Define

$$m_\rho \equiv m(R_\rho) = \big(\rho, g_A(\rho), 1, \theta_1, 1, \theta_1\big)^\top.$$

Fix any $h \in \mathcal{H}_{\mathrm{QC}}$, and write $d \equiv \nabla h(m_0) = (d_1, \ldots, d_6)^\top$, which exists due to the continuous differentiability of h on U. By the consistency condition on h, we have $h(m(R_\rho)) = \rho$ for $\rho \in (-1,1)$. Differentiating this identity at $\rho = 0$ gives $d_1 + \eta d_2 = 1$, where $\eta \equiv g_A'(0) = 4c'(0) - 1 = 1 - 8/\pi^2$.

In the conditional CLT of Corollary S1 and using the scale-invariance of h, the influence variable of h at $\rho_\tau = 0$ is $\psi_h(0) = d^\top\big(\zeta_{X,1}(R_0) - m_0\big)$, such that $\widetilde{V}_{\mathrm{QC}}(h;0) = \mathrm{Var}(\psi_h(0))$, which takes the explicit form $\psi_h(0) = d_1 T_C + d_2 T_A + D_h$, where $T_C \equiv \zeta_{C,1}^{(1)}\zeta_{C,1}^{(2)}$, $T_A \equiv \zeta_{A,1}^{(1)}\zeta_{A,1}^{(2)}$, and

$$D_h = d_3\big((\zeta_{C,1}^{(1)})^2 - 1\big) + d_4\big((\zeta_{A,1}^{(1)})^2 - \theta_1\big) + d_5\big((\zeta_{C,1}^{(2)})^2 - 1\big) + d_6\big((\zeta_{A,1}^{(2)})^2 - \theta_1\big).$$

At $\rho = 0$, the two Brownian paths are independent. Hence $\mathrm{Var}(T_C) = 1$, $\mathrm{Var}(T_A) = \theta_1^2$, and $\mathrm{Cov}(T_C, T_A) = 0$. Also, D_h is uncorrelated with T_C and T_A, because each term in D_h depends on only one Brownian path and $\mathbb{E}[\zeta_C^{(j)}] = \mathbb{E}[\zeta_A^{(j)}] = 0$ for $j \in \{1,2\}$. Therefore,

$$\widetilde{V}_{\mathrm{QC}}(h;0) = \mathrm{Var}\big(\psi_h(0)\big) = d_1^2 + \theta_1^2 d_2^2 + \mathrm{Var}(D_h) \geq d_1^2 + \theta_1^2 d_2^2.$$

Since $d_1 + \eta d_2 = 1$, the Cauchy–Schwarz inequality yields

$$1 = (d_1 + \eta d_2)^2 \le (d_1^2 + \theta_1^2 d_2^2)\Big(1 + \frac{\eta^2}{\theta_1^2}\Big).$$

Thus, for every $h \in \mathcal{H}_{\mathrm{QC}}$, we have

$$\widetilde{V}_{\mathrm{QC}}(h;0) \ge d_1^2 + \theta_1^2 d_2^2 \ge \frac{\theta_1^2}{\theta_1^2 + \eta^2},$$

which further implies that

$$\inf_{h\in\mathcal{H}_{\mathrm{QC}}} \sup_{\rho\in(-1,1)} \widetilde{V}_{\mathrm{QC}}(h;\rho) \ge \inf_{h\in\mathcal{H}_{\mathrm{QC}}} \widetilde{V}_{\mathrm{QC}}(h;0) \ge \frac{\theta_1^2}{\theta_1^2 + \eta^2}. \tag{SB.23}$$

We now show that the lower bound is attained by the proposed estimator $\hat{\rho}_n^*$. We first verify rigorously that the class of estimator $\hat{\rho}_n(\lambda) \in \mathcal{E}_{\mathrm{QC}}$ for all $\lambda \in [0,1]$. For each $\lambda \in [0,1]$, define the mapping

$$\Phi_\lambda(x) \equiv \frac{(1-\lambda)x_1 + \lambda x_2}{\sqrt{\{(1-\lambda)x_3 + \lambda x_4\}\{(1-\lambda)x_5 + \lambda x_6\}}}.$$

It is clear that $\Phi_\lambda(x) \in (-1,1)$ for all $x \in U$ and is continuously differentiable on U. Set $h_\lambda(x) = (g_\lambda^{-1} \circ \Phi_\lambda)(x)$, so that $\hat{\rho}_n(h_\lambda) \equiv \hat{\rho}_n(\lambda)$ for $\lambda \in [0,1]$. By Lemma 2, the continuous differentiability of g_λ^{-1} on $(-1,1)$ further implies that h_λ is continuously differentiable on U. The functional form of Φ_λ also ensures that h_λ is scale-invariant on U. Moreover, for any $\Sigma \in S_{++}^2$ with the decomposition $\Sigma = D^{1/2} R_\rho D^{1/2}$, we have

$$\Phi_\lambda\big(m(\Sigma)\big) = \Phi_\lambda\big(m(R_\rho)\big) = g_\lambda(\rho).$$

Therefore, $h_\lambda(m(\Sigma)) = (g_\lambda^{-1} \circ \Phi_\lambda)(m(\Sigma)) = (g_\lambda^{-1} \circ g_\lambda)(\rho) = \rho$, so h_λ satisfies the consistency condition. As a result, $h_\lambda \in \mathcal{H}_{\mathrm{QC}}$, and hence $\hat{\rho}_n(\lambda) \in \mathcal{E}_{\mathrm{QC}}$, for each $\lambda \in [0,1]$.

At $\rho = 0$ and since $g_\lambda(0) = 0$, the influence variable of h_λ simplifies to:

$$\psi_{h_\lambda}(0) = \frac{\zeta_\lambda^{(1,2)}}{g_\lambda'(0)} = \frac{(1-\lambda)T_C + \lambda T_A}{(1-\lambda) + \lambda\eta} \equiv a_\lambda T_C + b_\lambda T_A,$$

satisfying $a_\lambda + \eta b_\lambda = 1$. Thus the coefficient ratio is $b_\lambda/a_\lambda = \lambda/(1-\lambda)$. In the quadratic lower bound, the corresponding coefficients satisfy $d_1 + \eta d_2 = 1$ and $d_2/d_1 = \eta/\theta_1^2$. Therefore, the one-parameter estimator $\hat{\rho}_n(\lambda)$ attaining the lower bound has λ^* that satisfies $\lambda^*/(1-\lambda^*) = \eta/\theta_1^2$, or equivalently $\lambda^* = \eta/(\eta + \theta_1^2)$. But this is simply the optimal λ^* for the class $\mathcal{E}_{\mathrm{L}}$, which can be readily seen from its closed-form formula in (13):

$$\lambda^* = \frac{4c'(0) - 1}{4c'(0) - 1 + (3 - 4\log 2)^2} \equiv \frac{\eta}{\eta + \theta_1^2}.$$

Therefore, for the choice $h_{\lambda^*} \in \mathcal{H}_{\mathrm{QC}}$, we have

$$\widetilde{V}_{\mathrm{QC}}(h_{\lambda^*};0) = \frac{\theta_1^2}{\theta_1^2 + \eta^2} = \widetilde{V}(\lambda^*;0).$$

By the assumed worst-case-at-zero condition,

$$\sup_{\rho\in(-1,1)} \widetilde{V}_{\mathrm{QC}}(h_{\lambda^*};\rho) = \sup_{\rho\in(-1,1)} \widetilde{V}(\lambda^*;\rho) = \widetilde{V}(\lambda^*;0).$$

Therefore,

$$\inf_{h\in\mathcal{H}_{\mathrm{QC}}} \sup_{\rho\in(-1,1)} \widetilde{V}_{\mathrm{QC}}(h;\rho) \le \sup_{\rho\in(-1,1)} \widetilde{V}_{\mathrm{QC}}(h_{\lambda^*};\rho) = \widetilde{V}(\lambda^*;0).$$

Combining this upper bound with the lower bound proved in (SB.23) gives

$$\inf_{h\in\mathcal{H}_{\mathrm{QC}}} \sup_{\rho\in(-1,1)} \widetilde{V}_{\mathrm{QC}}(h;\rho) = \sup_{\rho\in(-1,1)} \widetilde{V}_{\mathrm{QC}}(h_{\lambda^*};\rho) = \widetilde{V}(\lambda^*;0).$$

This completes the proof. □

SC Additional Simulation Results

SC.1 Additional simulation setting

This section presents additional simulation evidence supporting Section 3 of the main text. Tables S1 and S2 report the simulated bias, variance, and relative MSE of $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, and $\hat{\rho}_{P,n}$ under the balanced sampling design with $(\lambda^{(1)},\lambda^{(2)}) = (1000, 1000)$ and the extreme-asynchronicity design with $(\lambda^{(1)},\lambda^{(2)}) = (1000, 100)$, respectively, complementing Table 2 of the main text. The sign accuracies for $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, and $\hat{\rho}_{P,n}$ under the balanced sampling design with $(\lambda^{(1)},\lambda^{(2)}) = (1000, 1000)$ and the extreme-asynchronicity design with $(\lambda^{(1)},\lambda^{(2)}) = (1000, 100)$ are presented in Figure S2, complementing Figure 5 of the main text. The finite sample coverage rates for the 95% fixed-k and large-k confidence intervals of $\hat{\rho}_n^*$ under $(\lambda^{(1)},\lambda^{(2)}) \in \{(1000, 1000), (1000, 100)\}$ are presented in Figure S3, complementing Figure 6 of the main text.

	Bias			Variance ×100			Relative MSE	
	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{P,n}$
Panel 1: $\gamma = 0, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 1000)$								
$k = 5$	-0.023	-0.038	-0.048	9.704	15.392	10.252	0.538	0.596
$k = 10$	-0.011	-0.017	-0.023	6.248	8.693	6.454	0.571	0.611
$k = 30$	-0.005	-0.006	-0.009	4.099	4.841	4.150	0.596	0.625
$k = 60$	-0.003	-0.004	-0.006	3.559	3.900	3.583	0.612	0.633
Panel 2: $\gamma = 0.5, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 1000)$								
$k = 5$	-0.022	-0.036	-0.048	9.699	15.742	10.365	0.525	0.596
$k = 10$	-0.013	-0.019	-0.026	6.181	8.769	6.383	0.552	0.599
$k = 30$	-0.005	-0.007	-0.010	4.021	4.792	4.067	0.595	0.619
$k = 60$	-0.004	-0.004	-0.006	3.493	3.884	3.517	0.606	0.630
Panel 3: $\gamma = 1, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 1000)$								
$k = 5$	-0.023	-0.038	-0.048	9.898	16.033	10.451	0.533	0.594
$k = 10$	-0.014	-0.020	-0.026	6.194	8.789	6.367	0.563	0.604
$k = 30$	-0.008	-0.007	-0.012	4.019	4.818	4.061	0.598	0.629
$k = 60$	-0.006	-0.004	-0.008	3.457	3.855	3.473	0.614	0.636
Panel 4: $\gamma = 2, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 1000)$								
$k = 5$	-0.031	-0.040	-0.055	9.738	15.468	10.241	0.549	0.613
$k = 10$	-0.021	-0.024	-0.035	6.265	9.053	6.531	0.563	0.619
$k = 30$	-0.013	-0.008	-0.018	3.938	4.785	4.000	0.597	0.628
$k = 60$	-0.012	-0.006	-0.014	3.393	3.846	3.433	0.612	0.641

Table S1: Finite-sample bias, variance, and relative mean squared error of $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, and $\hat{\rho}_{P,n}$ under the stochastic correlation model with balanced sampling design and measurement error. For each estimator, the relative MSE is calculated relative to that of the benchmark return-based estimator $\hat{\rho}_{C,n}$. All moments are simulated based on 10,000 Monte Carlo realizations. The parameter γ controls the noise-to-signal ratio of the measurement error, while $(\lambda^{(1)}, \lambda^{(2)})$ denotes the per-minute average number of Poisson observation arrivals.

	Bias			Variance ×100			Relative MSE	
	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{P,n}$
Panel 1: $\gamma = 0, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 100)$								
$k = 5$	-0.030	-0.037	-0.056	9.821	15.544	10.410	0.548	0.617
$k = 10$	-0.019	-0.020	-0.033	6.304	8.894	6.533	0.573	0.622
$k = 30$	-0.014	-0.009	-0.020	4.102	4.866	4.172	0.619	0.660
$k = 60$	-0.013	-0.008	-0.016	3.542	3.942	3.570	0.627	0.658
Panel 2: $\gamma = 0.5, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 100)$								
$k = 5$	-0.035	-0.038	-0.060	9.952	15.580	10.533	0.562	0.637
$k = 10$	-0.025	-0.022	-0.038	6.172	8.637	6.419	0.588	0.655
$k = 30$	-0.019	-0.011	-0.024	3.965	4.780	4.021	0.614	0.655
$k = 60$	-0.017	-0.008	-0.020	3.439	3.863	3.463	0.638	0.670
Panel 3: $\gamma = 1, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 100)$								
$k = 5$	-0.039	-0.040	-0.066	9.472	15.473	10.066	0.551	0.624
$k = 10$	-0.034	-0.028	-0.048	6.173	8.904	6.406	0.589	0.646
$k = 30$	-0.029	-0.016	-0.035	3.917	4.749	3.994	0.656	0.707
$k = 60$	-0.028	-0.015	-0.031	3.332	3.827	3.373	0.693	0.735
Panel 4: $\gamma = 2, (\lambda^{(1)}, \lambda^{(2)}) = (1000, 100)$								
$k = 5$	-0.077	-0.057	-0.101	10.073	15.828	10.483	0.618	0.690
$k = 10$	-0.069	-0.046	-0.081	6.130	9.150	6.405	0.658	0.730
$k = 30$	-0.064	-0.033	-0.066	3.595	4.666	3.684	0.813	0.853
$k = 60$	-0.063	-0.030	-0.063	3.011	3.714	3.081	0.985	1.002

Table S2: Finite-sample bias, variance, and relative mean squared error of $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, and $\hat{\rho}_{P,n}$ under the stochastic correlation model with extreme observation asynchronicity and measurement error. For each estimator, the relative MSE is calculated relative to that of the benchmark return-based estimator $\hat{\rho}_{C,n}$. All moments are simulated based on 10,000 Monte Carlo realizations. The parameter γ controls the noise-to-signal ratio of the measurement error, while $(\lambda^{(1)}, \lambda^{(2)})$ denotes the per-minute average number of Poisson observation arrivals.

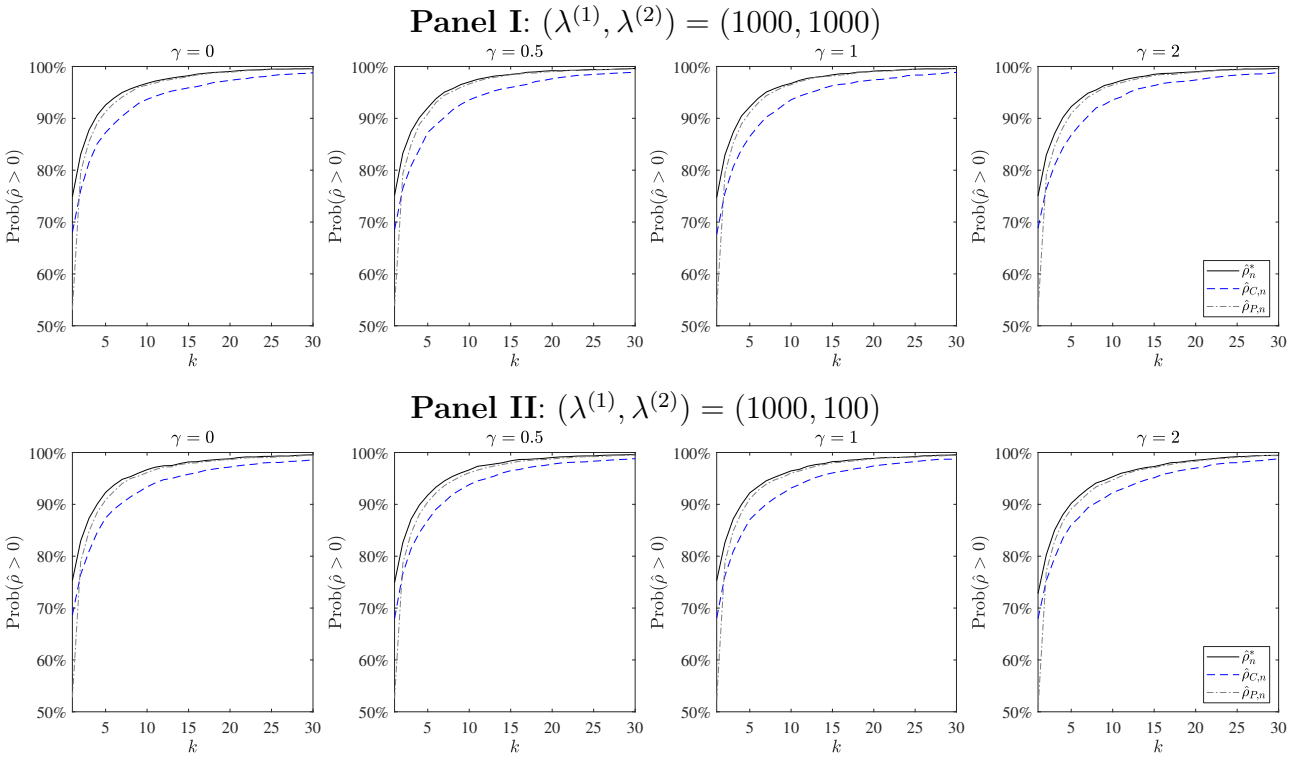


Figure S2: Finite-sample sign accuracies of $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, and $\hat{\rho}_{P,n}$ under the stochastic correlation model with the balanced setting $(\lambda^{(1)}, \lambda^{(2)}) = (1000, 1000)$ and the extreme asynchronicity setting $(\lambda^{(1)}, \lambda^{(2)}) = (1000, 100)$. The parameter γ controls the noise-to-signal ratio of the measurement error. The vertical axis reports the simulated probability that each estimator correctly identifies the sign of $\rho_0 > 0$.

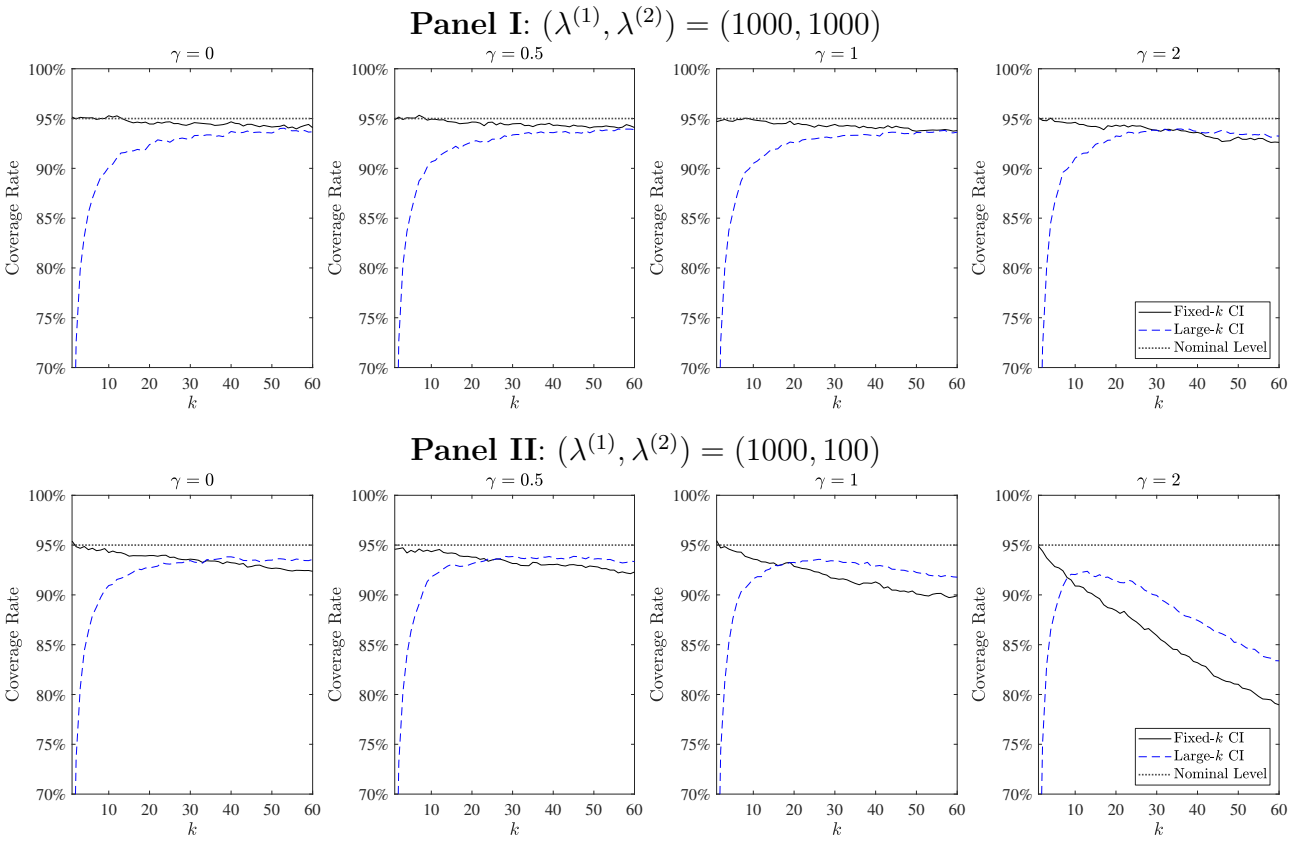


Figure S3: Finite-sample coverage rates of 95%-level large-k and fixed-k confidence intervals under the stochastic correlation model with the balanced setting $(\lambda^{(1)}, \lambda^{(2)}) = (1000, 1000)$ and the extreme asynchronicity setting $(\lambda^{(1)}, \lambda^{(2)}) = (1000, 100)$. The parameter γ controls the noise-to-signal ratio of the measurement error.

SC.2 Jump-robustness of candlestick-based correlation estimators

Under Assumption 1, the probability of a price jump occurring within the shrinking estimation window $[0, k_n\Delta_n]$ tends to zero as $k_n\Delta_n \to 0$. Consequently, jumps do not alter the asymptotic conclusions of Theorem 1. This observation motivates the continuous-price baseline simulations in Section 3, which isolate the effects of asynchronous observations and measurement error. However, the same argument does not characterize performance conditional on a jump occurring within the estimation window. We therefore conduct additional experiments in which every estimation window contains price jumps.

Specifically, we augment the continuous price processes in Section 3 as follows:

$$P_t^{(j)} + \mathbb{1}_{\{t \geq \tau^{(j)}\}} \xi^{(j)}, \qquad j \in \{1, 2\}.$$

For each asset, the jump time $\tau^{(j)}$ is uniformly distributed on $[0, \Delta_n]$, so that exactly one jump occurs within its first candle. The jump-size vector is bivariate Gaussian with

$$\begin{pmatrix} \xi^{(1)} \\ \xi^{(2)} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \nu^J \begin{pmatrix} 1 & \rho^J \\ \rho^J & 1 \end{pmatrix} \right),$$

where ν^J controls the marginal jump-size variance and ρ^J controls the correlation between jump sizes. The jump times and sizes are simulated independently of the continuous price process P. This design deliberately forces jumps into the estimation window and should be interpreted as a conditional simulation experiment.

We consider two specifications. In the independent-jump specification, $\tau^{(1)}$ and $\tau^{(2)}$ are independent and $\rho^J = 0$. In the co-jump specification, the assets share a common jump time, $\tau^{(1)} = \tau^{(2)}$, and $\rho^J = 0.5$. The common arrival time distinguishes co-jumps from the independent-jump design, while ρ^J governs dependence between their sizes. This specification is motivated by the literature on common price jumps (Bollerslev et al., 2008; Jacod and Todorov, 2009; Caporin et al., 2017).

We vary $\nu^J \in \{0.1, 0.5, 1\}$ and retain the stochastic correlation model, with noise-to-signal ratio $\gamma = 0.5$ and observation-arrival intensities $(\lambda^{(1)}, \lambda^{(2)}) = (1000, 500)$. Applying the same observation and measurement-error schemes as in Section 3, we construct 60 consecutive pairs of candlesticks from the observed jump-augmented prices over $[0, 60\Delta_n]$. We compute $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, and $\hat{\rho}_{P,n}$ from the first $k \in \{5, 10, 30, 60\}$ pairs. Every estimate therefore includes the first candle containing a jump in each asset.

For comparison, we also compute the pure asymmetry estimator, $\hat{\rho}_{A,n} \equiv \hat{\rho}_n(1)$. Its robustness to large jumps follows from the construction of the asymmetry feature $A = H + L - C$ as the difference between the upper and lower wick lengths. The sum of these lengths equals the high-low range minus the absolute open-close return. For a candle containing a single jump, the asymmetry is bounded in magnitude by the range plus the absolute return of the underlying

continuous-price path, independently of the jump size. This is closely related to the range–return cancellation underlying the wick-based volatility estimator of Li et al. (2025). It limits the influence of jump magnitudes on $\hat{\rho}_{A,n}$ and allows the contribution of the affected candle to diminish as more candles are included.

Tables S3 and S4 report the bias, variance, and MSE relative to $\hat{\rho}_{C,n}$ under the independent-jump and co-jump specifications, respectively, based on 10,000 Monte Carlo replications.

Table S3 shows that the pure asymmetry estimator $\hat{\rho}_{A,n}$ has the smallest absolute bias and MSE under independent jumps. The Popov estimator outperforms $\hat{\rho}_n^*$, which retains an advantage over the return-based benchmark. Larger jumps increase the absolute bias and variance of estimators retaining return information, while $\hat{\rho}_{A,n}$ remains comparatively stable. Increasing k reduces absolute bias and variance for all four estimators.

Table S4 exhibits broadly similar patterns in variance and MSE to those in Table S3. The main difference concerns bias: co-jumps produce smaller negative biases, and absolute bias need not increase monotonically with jump size. This is consistent with positive jump dependence partially offsetting the attenuation in return-based correlation estimates.

These findings complement the efficiency results in the main text. Popov's averaging of separately computed correlations preserves a fixed contribution from asymmetry information, whereas $\hat{\rho}_n^*$ combines moments before normalization and can therefore remain dominated by large returns. This helps explain their reversed MSE ranking in windows containing jumps. Such conditional performance is compatible with the minimax property of $\hat{\rho}_n^*$ in the local continuous-price limiting experiment and supports considering $\hat{\rho}_{A,n}$ when the estimation window is known to contain jumps.

	Bias				Variance ×100				Relative MSE		
	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_{A,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_{A,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{P,n}$	$\hat{\rho}_{A,n}$
Panel I: $\nu^J = 0.1$											
$k = 5$	-0.383	-0.415	-0.271	-0.126	47.650	58.852	20.746	22.179	0.818	0.360	0.280
$k = 10$	-0.329	-0.367	-0.220	-0.073	35.984	45.261	14.862	11.994	0.793	0.318	0.167
$k = 30$	-0.230	-0.277	-0.152	-0.028	20.185	26.957	8.852	5.535	0.720	0.284	0.083
$k = 60$	-0.172	-0.218	-0.117	-0.017	12.918	18.003	6.457	4.207	0.663	0.276	0.061
Panel II: $\nu^J = 0.5$											
$k = 5$	-0.480	-0.495	-0.311	-0.126	69.679	77.079	25.017	21.895	0.913	0.337	0.206
$k = 10$	-0.448	-0.471	-0.273	-0.075	60.235	67.528	20.208	12.107	0.895	0.301	0.111
$k = 30$	-0.379	-0.414	-0.222	-0.030	43.542	50.871	14.480	5.596	0.849	0.272	0.042
$k = 60$	-0.323	-0.366	-0.192	-0.018	32.849	40.012	11.527	4.272	0.804	0.264	0.025
Panel III: $\nu^J = 1$											
$k = 5$	-0.500	-0.511	-0.320	-0.129	76.740	82.736	26.713	22.350	0.936	0.335	0.196
$k = 10$	-0.478	-0.494	-0.286	-0.078	68.897	74.944	22.125	12.451	0.924	0.298	0.102
$k = 30$	-0.425	-0.452	-0.243	-0.034	53.817	60.562	16.746	5.744	0.887	0.269	0.036
$k = 60$	-0.379	-0.414	-0.217	-0.021	43.145	50.232	13.878	4.299	0.852	0.262	0.021

Table S3: Finite-sample bias, variance, and relative mean squared error of $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, $\hat{\rho}_{P,n}$, and $\hat{\rho}_{A,n}$ under the stochastic correlation model with independent price jumps, moderate observation asynchronicity $(\lambda^{(1)}, \lambda^{(2)}) = (1000, 500)$ and measurement error noise-to-signal ratio $\gamma = 0.5$. For each estimator, the relative MSE is calculated relative to that of the benchmark return-based estimator $\hat{\rho}_{C,n}$. All moments are simulated based on 10,000 Monte Carlo realizations. The parameter ν^J controls the variance of the independent jump size variable $(\xi^{(1)}, \xi^{(2)})$.

	Bias				Variance ×100				Relative MSE		
	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_{A,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{C,n}$	$\hat{\rho}_{P,n}$	$\hat{\rho}_{A,n}$	$\hat{\rho}_n^*$	$\hat{\rho}_{P,n}$	$\hat{\rho}_{A,n}$
Panel I: $\nu^J = 0.1$											
$k = 5$	-0.136	-0.157	-0.119	-0.081	38.866	48.874	20.145	20.682	0.790	0.400	0.357
$k = 10$	-0.107	-0.129	-0.087	-0.045	27.575	35.669	13.545	11.612	0.761	0.352	0.238
$k = 30$	-0.067	-0.089	-0.054	-0.020	14.078	19.468	7.303	5.594	0.690	0.307	0.137
$k = 60$	-0.047	-0.067	-0.040	-0.013	8.710	12.387	5.147	4.243	0.638	0.292	0.108
Panel II: $\nu^J = 0.5$											
$k = 5$	-0.176	-0.189	-0.129	-0.070	59.004	66.442	24.807	20.597	0.888	0.366	0.257
$k = 10$	-0.156	-0.173	-0.106	-0.038	49.066	56.401	18.811	11.465	0.867	0.319	0.143
$k = 30$	-0.118	-0.139	-0.079	-0.019	32.524	39.661	12.147	5.777	0.812	0.280	0.066
$k = 60$	-0.093	-0.116	-0.064	-0.011	22.939	29.482	9.133	4.284	0.763	0.270	0.042
Panel III: $\nu^J = 1$											
$k = 5$	-0.160	-0.168	-0.122	-0.075	63.993	69.966	25.597	20.698	0.916	0.361	0.250
$k = 10$	-0.147	-0.157	-0.101	-0.045	55.736	61.825	20.081	11.449	0.901	0.316	0.135
$k = 30$	-0.118	-0.133	-0.076	-0.019	40.635	47.238	13.849	5.456	0.857	0.276	0.053
$k = 60$	-0.098	-0.115	-0.064	-0.013	30.871	37.433	10.897	4.152	0.817	0.266	0.032

Table S4: Finite-sample bias, variance, and relative mean squared error of $\hat{\rho}_n^*$, $\hat{\rho}_{C,n}$, $\hat{\rho}_{P,n}$, and $\hat{\rho}_{A,n}$ under the stochastic correlation model with price co-jumps, moderate observation asynchronicity $(\lambda^{(1)}, \lambda^{(2)}) = (1000, 500)$ and measurement error noise-to-signal ratio $\gamma = 0.5$. For each estimator, the relative MSE is calculated relative to that of the benchmark return-based estimator $\hat{\rho}_{C,n}$. All moments are simulated based on 10,000 Monte Carlo realizations. The parameter ν^J controls the variance of the co-jump size variable $(\xi^{(1)}, \xi^{(2)})$ with $\text{Corr}(\xi^{(1)}, \xi^{(2)}) = 0.5$.

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